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THE PEOPLE'S
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MMPC-019
TOTAL QUALITY MANAGEMENT

Block

3

TOOLS AND TECHNIQUES

UNIT 7 **Statistical Quality Control**	**155**
UNIT 8 **Tools and Techniques of TQM**	**177**

BLOCK 3 TOOLS &TECHNIQUES

This block has the following two units and presents basic tools, techniques and concepts useful for developing and implementing a TQM programme. In addition to statistical quality control (SQC), Pareto Analysis and Six Sigma, several other concepts, tools and techniques e.g. Quality Circles, Benchmarking, Quality Function Deployment, Reengineering, Zero Defects etc. are also discussed.

Unit 7 **Statistical Quality Control**

Unit 8 **Tools and Techniques of TQM**

UNIT 7 STATISTICAL QUALITY CONTROL

Objectives

After reading this unit you should be able to:

- Understand the use of statistical thinking to quality improvement;
- Use seven tools to troubleshoot quality problems;
- Determine process capability of a manufacturing process;
- Explain the concept of a control chart;
- Understand the principles behind sampling methods.

Structure

7.1 Introduction to Statistical Quality Control

7.2 Process Capability

7.3 Seven Quality Improvement Tools

7.4 Control Charts

7.5 Control Charts for Attributes

7.6 Choosing the Correct SPC Chart

7.7 Acceptance Sampling

7.8 Summary

7.9 Key Words

7.10 Self-Assessment Questions

7.11 Further Readings

7.1 INTRODUCTION TO STATISTICAL QUALITY CONTROL

The concept of TQM is basically very simple. Each part of the organization has customers, some external and many internal. Identifying what the customer requirements are and setting about to meet them is the core of a total quality approach. This requires a good management system, methods including Statistical Quality Control (SQC), and teamwork.

A well-operated, documented management system provides the necessary foundation for the successful application of SQC. Note, however, that SQC is not just a collection of techniques. It is a strategy for reducing variability, the root cause of many quality problems. SQC refers to the use of statistical methods to improve or enhance quality for customer satisfaction. However, this task is seldom trivial because real-world processes are affected by numerous uncontrolled factors. For instance, within every factory, conditions fluctuate with time. Variations occur in the incoming materials, in machine conditions, in the environment, and in operator performance. A steel plant, for example, may purchase good quality ore from a mine, but the physical and chemical characteristics of ore coming from different locations in the mine may vary. Thus, everything isn't always "in control."

For e.g. in welding, it is not possible to form two exactly identical joints, and faulty joints may occur occasionally. In a cutting process, the size of each piece of material cut varies; even the most high-quality cutting machine has some inherent variability. In addition to such inherent variability, a large number of other factors may also influence processes. Many of these variations cannot be predicted with certainty, although sometimes it is possible to trace the unusual patterns of such variations to their root cause(s). If we have collected sufficient data from these variations, we can tell, in terms of probability, what is most likely to occur next if no action is taken. It we know what is likely to occur the next given certain conditions, we can take suitable actions to try to maintain or improve the acceptability of the output. This is the rationale of statistical quality control.

Another prospect in which statistical methods can help to improve product quality is the design of products and processes. It is now well-understood that over 2/3rd of all product malfunctions may be traced to their design. Indeed, the characteristics or quality of a product depends greatly on the choice of materials, settings of various parameters in the design of the product and the production process settings. In order to locate an optimal setting of the various parameters which gives the best product, we may consider using models governing the outcome and the various parameters, if such models can be established by theory or through experimental work.

The following two examples illustrate such cases.

Example 1: In the bakery industry, the taste, tenderness, and texture of a kind of bread depend on various input parameters, such as the origin of the flour used, the amount of sugar, the amount of baking powder, the baking temperature profile and baking time, and the type of oven used. To improve the quality of the bread produced, the baker may use a model that relates the input parameters and the output quality of the bread. It is a formidable task to find theoretical models quantifying the taste, tenderness, and texture of the bread produced and relates these quantities to the various input parameters based on our present scientific knowledge. However, the baker can easily use statistical methods in regression analysis to establish empirical models and use them to locate an optimal setting of the input parameters.

Example 2: Sometimes there are great difficulties in solving an engineering problem using established theoretical models. The heat accumulated on a chip in an electronic circuit during normal operation will raise the temperature of the chip and shorten its life. To improve the quality of the circuit, the designer would like to optimize the design of the circuit so that the heat accumulated on the chip will not exceed a certain level. This heat accumulated can be expressed theoretically in terms of other parameters in the circuit using a complicated system of ten or more daunting partial differential equations. It is usually not possible to solve such a system analytically, and to solve it numerically using the computer also has computational difficulties. In this situation, a statistical methodology known as design of experiments (DOE) can be used to find an optimal design of the circuit without going through the complicated method of solving partial differential equations.

In other cases, control may need to be exercised online while the process is in progress based on how the process is performing to maintain product quality. Thus, in statistical quality control, problems are numerous and diverse.

SQC engages the following three methodologies:

Acceptance Sampling

This method is also called “sampling inspection.” When products require inspection but it is not feasible to inspect 100% of the products, samples of the product may be taken for inspection, and conclusions drawn using the results of inspecting the samples. This technique specifies how to draw samples from a population and what rules to use to determine the acceptability of the product being inspected.

Statistical Process Control (SPC)

Even in an apparently stable production process, products produced are subject to random variations. SPC aims at controlling the variability of process output using a device called the control chart. On a control chart, a certain characteristic of the product is plotted. Under normal conditions, these plotted points are expected to vary in a “usual way” on the chart. When abnormal points or patterns appear on the chart, it is a statistical indication that the process parameters or production conditions might have changed undesirably. At this point, an investigation is conducted to discover unusual abnormal conditions (e.g., tool breakdown, use of wrong raw material, temperature controller failure, etc.). Subsequently, corrective actions are taken to remove the abnormality. In addition to the use of control charts, SPC also monitors process capability as an indicator of the adequacy of the manufacturing process to meet customer requirements under routine operating conditions. In summary, SPC aims to maintain a stable, capable, and predictable process.

Note, however, that since SPC requires processes to display measurable variation, it is ineffective for quality levels approaching six-sigma, though it is quite effective for companies in the early stages of quality improvement efforts.

Design of Experiments

Trial and error can be used to run experiments in the design of products and the design of processes in order to find an optimal setting of the parameters so that products of good quality will be produced. However, performing experiments by trial and error unscientifically is frequently very inefficient in the search for an optimal solution. Application of the statistical methodology of “design of experiments” (DOE) can help us in performing such experiments scientifically and systematically. Additionally, such methods greatly reduce the total effort used in product or process development experiments, increasing at the same time the accuracy of the results. DOE forms an integral part of Taguchi methods, techniques that produce high-quality and robust product and process designs.

The Correct Use of Statistical Quality Control Methods

The production of a product typically progresses as indicated in the simplified flow diagram shown in Figure 7.1. In order to improve the quality of the tonal product, design of experiments (DOE) may be used in Step 1 and Step 2, acceptance sampling may be used in Step 3 and Step 5, and statistical process control(SPC) may be used in Step 4.

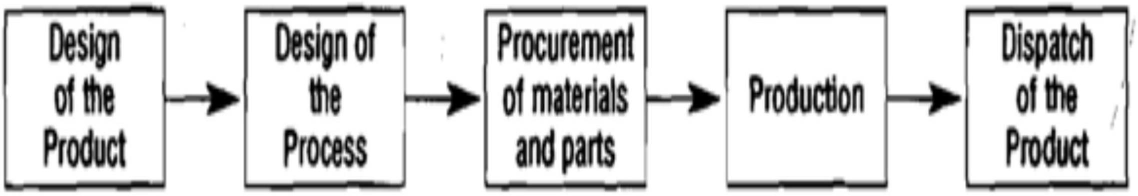

Figure 7.3: Production from Design to Dispatch

There are several benefits that the SPC approach brings which are as follows:

- There are no restrictions as to the type of the process being controlled or studied, but the process tackled will be improved.
- Decisions guided by SPC are based on facts not opinions. Thus a lot of emotion is removed from problems by SPC.
- Quality awareness of the workforce increases because they become directly involved in the improvement process.
- Knowledge and experience of those who operate the process are released in a systematic way through the investigative approach. They understand their role in problem solving, which includes collecting facts, communicating fads, and making decisions.
- Management and supervisors solve problems methodically, instead of by using a seat-of-the-pants style.

7.2 PROCESS CAPABILITY

We introduce now an important concept employed in thinking statistically about real life processes. Process capability is the range over which the "natural variation "of a process occurs as determined by the system of common or random causes; that is, process capability indicates what the process can deliver under "stable "conditions when it is said to be under statistical control.

The capability of a process is the fraction of output that can be routinely found to be within specifications (specs). A capable process has 99.73% or more of its output within specifications.

Process capability refers to a process's ability to produce parts within the range of engineering or customer specifications. Process control, on the other hand, refers to maintaining a process's performance at its current capability level. This involves activities such as sampling the process product, charting its performance, determining causes of excessive variation, and taking corrective action.

The capability of a process is an expression of how well the process can deliver output within the specifications compared to the range of natural variability seen in the process. Essentially, process capability expresses the proportion or fractional output that a process can routinely deliver within the specifications. By subjecting a process to a capability study, manufacturing and quality managers can determine whether the process needs improvement and to what extent.

Knowing process capability allows managers to quantitatively predict how well a process will meet specifications and specify the equipment requirements necessary to maintain its capability. For example, if a design specification requires metal tubing to be cut within one-tenth of an inch, a process consisting of a worker using a ruler and hacksaw will likely result in a large percentage of nonconforming product due to the high inherent or natural variability of the process. In such a case, the firm has three possible choices: (1) measure each piece and cut non-conforming tubing, (2) invest in new technology to develop a better process, or (3) change the specification.

Ultimately, such decisions are usually based on economics. It is important to note that the cost to produce one unit of a product remains the same under routine production, whether the product ultimately fails within or outside the specifications. As such, the firm may be forced to raise the market price of within-spec products, which may weaken its competitive position. Scrap and rework of nonconforming parts are therefore poor business strategies, as labour and materials have already been invested in producing the unacceptable product. Inspection errors may also allow some nonconforming products to leave the production facility if the firm aims to make parts that just meet the specifications. However, investing in new technology may require substantial investment that the firm cannot afford.

Changes in design, on the other hand, may sacrifice fitness-for-use requirements and result in a lower quality product. Thus, these factors demonstrate the need to consider process capability during product design and in the acceptance of new contracts. Many firms now require process capability data from their vendors. Both ISO 9000 and QS 9000 quality management systems require a firm to determine its process capability.

Process capability has three important components: (1) the design specifications, (2) the centering of the natural variation, and (3) the range or spread of variation. If the process is in control but cannot produce according to the design specifications, the question should be raised whether the specifications have been correctly applied or if they may be relaxed without adversely affecting the assembly or subsequent use of the product. If the specifications are realistic and firm, an effort must be made to improve the process to the point where it is capable of producing consistently within specifimcations.

Usually, this can be corrected by a simple adjustment of a machine setting or recalibrating the inspection equipment used to capture the measurements. If no action is taken, however, a substantial portion of output will fall outside the spec limits even though the process has the inherent capability to meet specifications.

We may define the study of process capability from another perspective. A capability study is a technique for analysing the random variability found in a production process. In every manufacturing process, there is some variability. This variability may be large or small, but it is always present. It can be divided into two types:

- Variability due to common (random) causes
- Variability due to assignable (special) causes

The first type of variability is said to be inherent in the process, and it can be expected to occur naturally within a process. It is attributed to a multitude of factors that behave like a constant system of changes affecting the process called common or random causes, Such factors include equipment vibration, passing traffic, atmospheric pressure or temperature changes, electrical voltage or humidity fluctuations, changes in the operator's physical or emotional conditions, etc. Such forces determine whether a coin when tossed will end up showing a head or tail when on the floor. Together, however, these "chances" form a unique, stable distribution.

Inherent variability may be reduced by changing the environment or the technology, but given a set of operating conditions, this variability can never be completely eliminated from a process. Variability due to assignable causes, on the other hand, refers to the variation that can be linked to specific or special causes that disturb a process. Examples are tool failure, power supply interruption, process controller malfunction, adding wrong ingredients or wrong quantities, switching a vendor, etc.

Assignable causes are fewer in number and are usually identifiable through investigation on the shop floor or an examination of process logs. The effect (i.e., the variation in the process) caused by an assignable factor, however, is usually large and detectable when compared with the inherent variability seen in the process. If the assignable causes are controlled properly, the total process variability associated with them can be reduced and even eliminated.

A capability study measures the inherent variability or the performance potential of a process when no assignable causes are present (i.e., when the process is said to be in statistical control). Since inherent variability can be described by a unique distribution, usually a normal distribution, capability can be evaluated by utilizing the properties of this distribution. Recall that capability is the proportion of routine process output that remains within product specs.

Even approximate capability calculations done using histograms enable manufacturers to take a preventive approach to defects. This approach is in contrast with the traditional two-step process: production personnel make the product while QC personnel inspects and screen out products that do not meet specifications. Such QC is wasteful and expensive since it allows plant resources including time and materials to be put into products that are not saleable. It is also unreliable since even 100 percent inspection would fail to catch all defective products. SPC aims to correct undesirable changes in the output of a process. Such changes may affect the centering (or accuracy) of the process, or its variability (spread or precision).

Control Limits are not an Indication of Capability

Those who are new to SPC often have the misconception that they don't need to calculate capability indices. Some even think that they can compare their control limits to the spec limits. This is not true because control limits look at the distribution of averages while capability indices look at the distribution of individual measurements. The statistical theory of the "central limit theorem" says that the averages of samples or subgroups follow a normal distribution more closely. This is why we can easily construct control charts on process data that are not normally distributed themselves. But averages cannot be used for capability calculation because capability evaluates individual parts delivered by a process. After all, parts get shipped to customers, not averages.

What Capability Studies Do for You

Capability studies are most often used to quickly determine whether a process can meet specs or how many parts will exceed the specifications. However, there are numerous other practical uses:

- Estimating the percentage of defective parts to be expected
- Evaluating new equipment purchases
- Predicting whether design tolerances can be met
- Assigning equipment to production
- Planning process control checks
- Analyzing the interrelationship of sequential processes
- Making adjustments during manufacture
- Setting specifications
- Costing out contracts
- Statistical Quality Control

Since a capability study determines the inherent reproducibility of parts created in a process, it can even be applied to many problems outside the domain of manufacturing, such as inspection, administration, and engineering.

There are instances where capability measurements are valuable even when it is not practical to determine in advance if the process is in control. Such an analysis is called a performance study. Performance studies can be useful for examining incoming lots of materials or one-time-only production runs. In the case of an incoming lot, a performance study cannot tell us that the process that produced the materials is in control, but it may tell us, by the shape of the distribution, what percent of the parts are out of specs or, more importantly, whether the distribution was truncated by the vendor sorting out the obvious bad parts.

How to Set Up a Capability Study

Before setting up a capability study, we must select the critical dimension or quality characteristic (it must be a measurable variable) to be examined. This dimension is the one that must meet product specs. In the simplest case, the study dimension is the result of a single, direct product and measurement process. In more complicated studies, the critical dimension may be the result of several processing steps or stages. It may become necessary in these cases to perform capability studies on each process stage. Studies on early process stages frequently prove to be more valuable than elaborate capability studies done on later processes since early processes lay the foundation (i.e., constitute the input) that may affect later operations.

Once the critical dimensions are selected, data measurements can be collected. This can be accomplished manually or by using automatic gauging and filtering linked to a data collection device or computer. When measurements on a critical dimension are made, it is important to ensure that the measuring instrument is as precise as possible, preferably one order of magnitude finer than the specification. Otherwise, the measuring process itself will contribute excess variation to the dimension data as recorded. Using handheld data collectors with automatic gauges may help reduce errors introduced by the process of measurement, data recording, and transcription for post-processing by computer.

The deal situation for data collection is to collect as much data as possible over a defined time period. This will yield reliable capability number since it is based upon a large sample size. In the practice of process improvement, determining process capability is Step 5.

Capability Calculations condition 1: The Process Must be in Control

Process capability formulas commonly used by industry require that the process must be in control and normally distributed before one takes samples to estimate process capability. All standard capability indices assume that the process is in control and the individual data follow a normal distribution. If the process is not in control, capability indices are not valid, even if they appear to indicate the process is capable.

Three different statistical tools are used together to determine whether a process is in control and follows a normal distribution. These are

- Control chart
- Visual analysis of a histogram
- Mathematical analysis of the distribution to test that the distribution is normal.

Note that no single tool can do the job here and all three must be used together. Control charts are the most common method for maintaining a process in statistical control. For a process to be in control, all points plotted, on the control chart must be inside the control limits with no apparent patterns (e.g., trends) be present. A histogram (allows us to quickly see (a) if any parts are outside the spec limits and (b) what the distribution's position is relative to the specification range. If the process is one that is naturally a normal distribution, then the histogram should approximate a bell-shaped curve if the process is in control. However, note that a process can be in control but not have its - individuals following a normal distribution if the process is inherently non-normal.

Capability Calculations condition 2: The Process Must be Inherently Normal

Many processes naturally follow a bell-shaped curve (a normal distribution) but some do not. Examples of non-normal dimensions are round, square, flat and positional tolerances; they have a natural barrier at zero. In these cases, a perfect measurement is zero.

There can never be a value less than zero. The standard capability indices are not valid for such non-normal distributions. Tests for normality are available in SPC text books that can assist you to identify whether or not a process is normal. If a process is not normal, you may have to use special capability measures that apply to non-normal distributions.

Statistical Process Control (SPC) Methodology

Control charts, like the other basic tools for quality improvement, are relatively simple to use. In general, control charts have two objectives, (a) help restore accuracy of the process so that the process average stays near the target, and (b) help minimize variation in the process to ensure that good precision is maintained in the output .

Control charts have three basic applications: (1) to establish a state of statistical control, (2) to monitor a process and signal when the process goes out of control, and (3) to determine process capability. The following is a summary of the steps required to develop and use control charts. Steps 1 through 4 focus on establishing a state of statistical control; in step 5, the charts are used for on-going monitoring; and finally in step 6, the data are used for process capability analysis.

1. Preparation

   a. Choose the variable or attribute to be measured

   b. Determine the basis, size and frequency of sampling.

   c. Set up the correct control chart.

2. Data Collection

   a. Record the data

   b. Calculate relevant statistics:

   c. Plot the statistics on the chart.

3. Determination of trial control limits.

   a. Draw the center line (process average) on the chart.

   b. Compute the upper and lower control limits.

4. Analysis and interpretation

   a. Investigate the chart for lack of control

   b. Eliminate out-of-control points.

   c. Re-compute control limits if necessary.

5. Use as a problem-solving tool

   a. Continue data collection and plotting.

   b. Identify out-of-control situations and take corrective action.

6. Use the control chart data to determine process capability, if desired.

7.3 SEVEN QUALITY IMPROVEMENT TOOLS

In SPC, numbers and information form the basis for decisions and actions. Therefore, a thorough data recording system bimanual or otherwise would be an essential enabler for SPC. In order to allow one to interpret fully and derive maximum use of quality-related data, over the past fifty years a set of simple statistical' tools' have evolved. These tools offer any organization an easy means to collect, present and analyse most of such data. In this section we briefly review these tools.

In the 1950's Japanese industry began to learn and apply statistical methods in earnestness, methods that American statisticians Walter Shewhart and W Edward Deming developed in the 1930's and 1940's to help manage quality. Subsequently, progress in continuous quality improvement in Japan led to significant expansion of the many simple statistical tools on shop floor. Kaoru Ishikawa, head of the Japanese Union of Scientists and Engineers (JUSE), later formalized the use of these tools in Japanese manufacturing with the introduction of the 7 Quality Control (7 QC) tools. The seven Ishikawa tools reviewed below are now an integral part of quality control on the shop floor around the world. Many Indian industries use them routinely. Some of these quality improvements tools were discussed earlier in the first Part.

Flowchart

The flowchart lists the order of activities in a project or process and their interdependency. It expresses detailed process knowledge. To express this knowledge certain standard symbols are used. The oval symbol indicates the beginning or end of the process. The boxes indicate action items while diamonds indicate decision or check points. The flowchart can be used to identify the steps affecting quality and the potential control points. Another effective use of the flowchart would be to map the ideal process and the actual process and to identify their differences as the targets for improvements. Flowcharting is often the first step in Business Process Reengineering (BPR).

Histogram

The histogram is a bar chart showing a distribution of variable quantities or characteristics. An example of a "live" histogram would be to line up by height a group of students enrolled in a course. Normally, one individual would be the tallest and one the shortest, with a cluster of individuals bunched around the average height.

In manufacturing, the histogram can rapidly identify the nature of quality problems in a process by the shape of the distribution as well as the width of the distribution. It informally establishes process capability. It can also help compare two or more distributions.

Pareto Chart

The Pareto chart, indicates the distribution of effects attributable to various causes or factors arranged from the most frequent to the least frequent. This tool is named after Wilfred Pareto, the Italian economist who determined that wealth is not evenly distributed and some of the people have most of the money.

This tool is a graphical picture of the relative frequencies of different types of quality problems with the most frequent problem type obtaining dear visibility. Thus the Pareto chart identifies the vital few and the trivial many and it highlights problems that should be worked first to get the most improvement. Historically,

Cause and Effect Diagram

The cause and effect diagram is also called the fishbone chart because of its appearance and the Ishikawa diagram after the man who popularized its use in Japan. Its most frequent use is to list the causes of some particular quality problem or defect. The lines coming of the core horizontal line are the main causes while the lines coming off those are sub causes.

The cause and effect diagram identifies problem areas where data should be collected and analysed. It is used to develop reaction plans to help investigate out-of-control points found on control charts. It is also the first step for planning design of experiments (DOE) studies and for applying Taguchi methods to improve product and process designs.

Scatter Diagram

The scatter diagram shows any existing pattern in the relationship between two variables that are thought to be related. For example, is there a relationship between outside temperature and cases of the common cold? As temperatures drop, do cases of the common cold rise in number? The closer the scatter points hug a diagonal line, the more closely there is one-to-one relationship between the variables being studied. Thus, the scatter diagram may be used to develop informal models to predict the future based on past correlations.

Run Chart

The run chart shows the history and pattern of variation. It is plot of data points in time sequence, connected by a line. Its primary use is in determining trends over time. The analyst should indicate on the chart whether up is good or down is good. This tool is used at the beginning of the change process to see what the problems are. It is also used at the end (or check) part of the change process to see whether the change made has resulted in a permanent process improvement.

Control Chart

Whereas a histogram gives a static picture of process variability, a run chart or a control chart illustrates the dynamic pet romance (i.e., performance over time) of the process. The control chart in particular is a powerful process quality monitoring device and it constitutes the core of statistical process control (SPC). It is a line chart marked with control limits at 3 standard deviations (s) above and below the average quality level.

These limits are based on the statistical studies of shop data conducted in the 1930s by Dr Walter Shewart.

Failure to distinguish between common causes and special causes of variation can actually increase the variation in the output of a process. This is often due to the mistaken belief that whenever process output is off target, some adjustment must be made. However, knowing when to leave a process alone is an important step in maintaining control over a process. Equally important is deciding when to take action to prevent the production of nonconforming product. Using actual industry data Shewhart demonstrated that a sensible strategy to control quality is to first eliminate the special causes with the help of the control and then systematically reduce the common causes. This strategy reduces the variation in process output with a high degree of reliability while it improves the acceptability of the product.

Statistical process control (SPC) is actually a methodology for monitoring a process to(a) identify the special causes of variation and (b) signal the need to take corrective action when it is appropriate. When special causes are present, the process is deemed to be out of control. If the variation in the process is due to common causes alone, the process is said to be in statistical control. A practical definition of statistical control is that both the process averages and variances are constant over time . Such a process is stable and predictable.

SPC uses control charts as the basic tool to improve both quality and productivity. SPC provides a means by which a firm may demonstrate its quality capability, an activity necessary for survival in today's highly competitive markets. Also, many customers (e.g., the automotive companies) now require the evidence that their suppliers use SPC in managing their operations. Note, however, that since SPC requires processes to display measurable variation; even though it is quite effective for companies in the early stages of quality efforts, it becomes ineffective in producing improvements once quality level approaches six-sigma.

7.4 CONTROL CHARTS

As we mentioned in the previous section, the control chart is a powerful process quality monitoring device and it constitutes the core of statistical process control (SPC). In SPC methodology knowing when to leave a process in itself is an important step in maintaining control over a process. Control charts enable us to do that.

Equally important is knowing when to take action to prevent the production of nonconforming product.

Indeed, failure to distinguish between variation produced by common causes and special causes can actually increase the variation in the output of a process. Again, control charts empower us here.

When the chart crosses any of its control limits, special causes are indicated to be present. The process is now deemed to be out of control and is investigated to remove the source of disturbance. Otherwise, when variation stays within control limits, it is indicated to be due to common causes only. Now the process is said to be in "statistical control" and it should be left alone. Statistical control is defined as the state in which both the process averages and variances are constant over time and hence the process output is stable and predictable. Control charts help us in bringing a process within such control.

Most processes that deliver a "product" or a "service" may be monitored by measuring their output over time and then plotting these measurements appropriately. However, processes differ in the nature of those outputs. Variables data are those output characteristics that are measurable along a continuous scale.

Examples of variables data are length, weight, or viscosity. By contrast, some output may only be judged to be good or bad, or "acceptable" or "unacceptable", such as print quality of a photocopier or defective knots produced per meter by a weaving machine. In such cases we categorize the output as an attribute that is either acceptable or unacceptable; we cannot put it on a continuous scale as done with weight or viscosity.

However, SPC methodology provides us with a variety of different types of control charts to work with such diversity.

For variables data control charts most commonly used are the "x˜" chart and the "R-chart" (range chart). The x chart is used to monitor the centring of the process to help control its accuracy . The R-chart monitors the dispersion or precision of the process.Range R rather than standard deviation is used as a measure of variation simply to enable workers on the factory floor perform control chart calculations by hand, as done for example in the turbine blade machining shop in BHEL, Hardwar. For large samples and when data can be processed by a computer. The standard deviation is a better measure of variability.

Interpreting Abnormal Patterns in Control Charts

When a process is in statistical control the points on a control chart fluctuate randomly between the control limits with no recognizable, non-random pattern. The following checklist provides a set of general rules tor examining a process to determine if it is in control:

1. No points are outside control limits.
2. The number of points above and below the center line is about the same.
3. The points seem to fall randomly above and below the center line.
4. Most points, but not all, are near the center line, and only a few are dose to the control limits.

The underlying assumption behind these rules is that the distribution of sample means is normal. This assumption follows from the central limit theorem of statistics, which states that the distribution of sample means approaches a normal distribution as the sample size increases regardless of the original distribution. Of course, for small sample sizes, the distribution of the original data must be reasonably normal for this assumption to hold. The upper and lower control limits are computed to be three standard deviations from the overall mean. Thus, probability that any sample mean falls outside the control limits is very small. This probability is the origin of rule 1.

Since the normal distribution is symmetric, about the same number of points fall above as below the centre line. Also, since the mean of the normal distribution is the median, about half the points fall on either side of the centre line. Finally, about 68 percent of a normal distribution falls within one standard deviation of the mean; thus, most but not all points should be close to the centre line, These characteristics will hold provided that the mean and variance of the original data have not changed during the time the data were collected; that is, the process is stable.

Several types of unusual patterns arise in control charts, which are reviewed here along with an indication of the typical causes of such patterns.

One Point Outside Control Limits

A single point outside the control limits is usually produced by a special cause. Often, the R-chart provides a similar indication. Once in awhile, however, such points are normal part of the process and occur simply by chance.

A common reason for a point falling outside a control limit is an error in the calculation of x bar or R for the sample. You should always check your calculations whenever this occurs, other possible cause area sudden power surge, a broken tool, measurement error, or an incomplete or omitted operation in the process.

Sudden Shift in the Process Average

An unusual number of consecutive points falling on one side of the centre line are usually an indication that the process average has suddenly shifted. Typically, this occurrence is the result of an external influence that has affected the process, which would be considered a special cause. In both the x bar and R-charts, possible causes might be a new operator, a new inspector, a new machine setting, or a change in the setup or method.

If the shift is up in the R-chart, the process has become less uniform. Typical causes are carelessness of operators, poor or inadequate maintenance, or possibly a fixture in need of repair. If the shift is down in the R-chart, the uniformity of the process has improved. This might be the result of improved workmanship or better machines or materials. As mentioned, every effort should be made to determine the reason for the improvement and to maintain it.

Three rules of thumb are used for early detection of process shifts. A simple rule is that if eight consecutive points fall on one side of the centre line, one could conclude that the mean has shifted. Second, divide the region between the centre line and each control limit into three equal parts. Then if (a) two of three consecutive points fall in the outer one-third region between the centre line and one of the control limits or (b) four of five consecutive points fall within the outer two-thirds region, one would also control that the process has gone out of control.

Cycles

Cycles are short, repeated patterns in the chart, alternating high peaks and low valleys These patterns are the result of causes that come and go on a regular basis. In the x bar chart, cycles maybe the result of operator rotation or fatigue at the end of a shift, different gauges used by different inspectors, seasonal effects such as temperature or humidity or differences between day and night shifts. In the R-chart cycles can occur from maintenance schedules, rotation of fixtures or gages, differences between shifts, or operator fatigue.

Trends

A trend is the result of some cause that gradually affects the quality characteristics of the product and causes the points on a control chart to gradually move up or down from the centre line. As a new group of operators gains experience on the job, for example, or as maintenance of equipment improves over time, a trend may occur. In the x bar chart, trends may be the result of improving operator skills, dirt or chip build-up in fixtures, tool wear, changes in temperature or humidity, or aging of equipment. In the R-chart, an increasing trend may be due to a gradual decline in material quality, operator fatigue, gradual loosening of a fixture or a tool, or dulling of a tool. A decreasing trend often is the result of improved operator skill or work methods, better purchased materials, or improved or more frequent maintenance.

Hugging the Centre Line

Hugging the centre line occurs when nearly all the points fall dose to the centre line In the control chart it appears that the control limits are too wide. A common cause of hugging the centre line is that the sample includes one item systematically taken from each of several machines, spindles, operators, and so on. A simple example will serve to illustrate this pattern.

Hugging the Control Limits

This pattern shows up when many points are near the control limits with very few in between. It is often, called a mixture and is actually a combination of two different patterns on the same chart. A mixture can be split into two separate patterns.

A mixture pattern can result when different lots of material are used in one process, or when parts are produced by different machines but fed into a common inspection group.

Instability

Instability is characterized by unnatural and erratic fluctuations on both sides of the chart over a period of time. Points will often lie outside both the upper and lower control limits without consistent pattern. Assignable causes may be more difficult to identify in this case than when specific patterns are present. A frequent cause of instability is over adjustment of a machine, or the same reasons that cause hugging the control limits.

As suggested earlier, the R-chart should be analysed before the x bar chart, because some out-of-control conditions in the R-chart may cause out-of-control conditions in the chart. Also, as the variability in I the process decreases, all the sample observations will be closer to the true population mean, and therefore their average, x bar, will not vary much from sample to sample. If this reduction in the variation can be identified and controlled, then new control limits should be computed for both charts.

Routine Process Monitoring and Control

After a process is determined to be in control, the charts should be used on a daily basis to monitor production, identify any special causes that might arise, and make corrections as necessary. More important, the chart tells when to leave the process alone. Unnecessary adjustments to a process result in non-productive labour, reduced production, and increased variability of output.

It is more productive if the operators themselves take the samples and chart the data. In this way, they can react quickly to changes in the process and immediately make adjustments. To do this effectively, training of the operators is essential. Many companies conduct in-house training programmes to teach operators and supervisors the elementary methods of statistical quality control. Not only does this training provide the mathematical and technical skills that are required, but it also give the shop floor personnel increased quality consciousness.

Introduction of control charts on the shop floor typically leads to improvements in conformance, particularly when the process is labour intensive.

Another important point must be noted. Control charts are designed to be used by production operators rather than by inspectors or QC personnel. Under the philosophy of statistical process control, the burden of quality rests with the operators themselves. The use of control charts allows operators to react quickly to special causes of variation. The range is used in place of the standard deviation for the very reason that it allows shop-floor personnel to easily make the necessary computations to plot points on a control chart. The right approach taken by management in ingraining the correct outlook among the workers appears to hold the key here.

Estimating Plant Process Capability

After a process has been brought to a state of statistical control by eliminating special causes of variation, the data may be used to find a rough estimate process capability. This approach uses the average range R bar rather than the estimated standard deviation of the original data. Nevertheless, it is a quick and useful method, provided that the distribution of the original data is reasonably normal.

Use of Warning Limits

When a process is in control, the x bar and R charts may be used to make decisions about the state of the process during its operation. We use three zones on the charts to help in the routine management of the process. The zones are marked on the charts as follows.

Zone 1 (it falls between the upper and lower WARNING LINES (or ± 2 sigma lines). If the plotted points fall in this zone, it indicates that the process has remained stable and actions or adjustments are unnecessary. Indeed any adjustment here may increase the amount of variability.

Zone 2 (it falls between Zone 1 and Zone 3). Any point found in Zone 2 suggests that there may have been an assignable change and another sample must be taken to check that out.

Zone 3 (it falls beyond the upper or Lower Control Limit). Any points falling in this zone indicate that the process should be investigated and that if action is taken, the latest estimate of x-double bar value and R bar should be used to revise the control limits.

7.5 CONTROL CHARTS FOR ATTRIBUTES

Attributes quality data assume only two values of good or bad, pass or fail. Attributes usually cannot be measured, but they can be observed and counted and are useful in quality management in many practical situations. For instance, in printing packages for consumer products, colour quality can be rated as acceptable or not acceptable, or a sheet of cardboard is either damaged or is not. Usually, attributes data are easy to collect, often by visual inspection. Many accounting records, such as percent scrapped, are also usually readily available. However, one drawback in using attributes data is that large samples are necessary to obtain valid statistical results.

7.6 CHOOSING THE CORRECT SPC CHART

Confusion often exists over which chart is appropriate for a specific application, since the c- and u-charts apply to situations in which the quality characteristics inspected do not necessarily come from discrete units.

The key issue to consider is whether the sampling unit is constant. For example, suppose that an electronics manufacturer produces circuit boards. The boards may contain various defects, such as faulty components and missing connections. Because the sampling unit - the circuit board - is constant (assuming that all boards are the same), a c-chart is appropriate. If the process produces boards of varying sizes with different numbers of components and connections, then a u-chart would apply.

As another example, consider a telemarketing firm that wants to track the number of calls needed to make one sale. In this case, the firm has no physical sampling unit. However, an analogy can be made with the circuit boards. The sale corresponds to the circuit board, and the number of calls to the number of defects. In both examples, the number of occurrences in relationship to a constant entity is being measured. Thus a c-chart is appreciated.

7.7 ACCEPTANCE SAMPLING

Acceptance sampling is a method used to distinguish between acceptable and poor quality products, but it does not actually improve the quality of the products. If it were possible to inspect or use every product, a 100% inspection would be performed, and acceptance sampling would not be necessary. However, 100%

inspection is often not feasible or cost-effective, particularly for low-cost mass-produced items such as injection-moulded parts or flash light bulbs, where the cost of inspection may exceed the value of the product. In such cases, acceptance sampling can be used, where a small fraction of the product is inspected using a sampling plan.

A sampling plan outlines the procedure for drawing a sample from a batch of products and deciding whether to accept or reject the entire batch based on the results of the sample inspection. The sample size is typically smaller than the whole batch, and rejection of the batch may involve downgrading it, selling it at a lower price, or returning it to the supplier.

For example, a sampling plan may specify that a random sample of n items is drawn from a batch, and the batch is rejected only if c or more of these n items are defective or non-conforming. The operating characteristic curve (OC-curve) of a sampling plan is a graph that shows the probability of accepting the batch (Pa(p)) as a function of the fraction p of defective products in the batch. A higher Pa(p) benefits the producer, while a smaller Pa(p) benefits the consumer who seeks assurance against receiving poor quality products and prefers to accept batches with lower Pa(p) (fraction defective) values.

AQL and BQL

To specify a sampling plan with the desired Pa(p) characteristics, the values of n and c must be appropriately determined. This determination involves specifying two key quality parameters: the AQL (acceptable quality level) and the RQL (rejection quality level). The AQL represents the maximum acceptable level of defects in a batch, while the RQL represents the minimum level of defects that would result in a batch being rejected.

A perfect quality batch with no defective items is often considered unattainable. Therefore, customers typically tolerate a small fraction of defective items in their purchases, as long as the fraction does not exceed the AQL. However, customers also require a high level of assurance that the sampling plan will reject batches with fractions of defective items that exceed a poor quality threshold, represented by the RQL.

The RQL is the defect level that would cause significant issues once a lot enters the customer's factory. In contrast, the supplier of the parts aims to ensure that the customer's acceptance sampling plan will not reject too many batches that fall within the customer's acceptable defect level, represented by the AQL. The AQL represents the worst fraction of defective items that the customer can accept on average in the shipments they receive.

The Operating Characteristic (OC) Curve of a Sampling Plan

A bit of thinking will indicate that only error-free 100% inspection of all items in a lot would accept the batch with 100% probability if the fraction of defective items in it is less than or equal to AQL and reject the lot with a 100% probability if the fraction of defective items in it is larger than AQL. Such performance cannot be realized otherwise.

In order to correctly evaluate the OC-curve of an arbitrary (not 100% inspection) sampling plan with parameters (n, c), we need to use certain principles of probability theory. Suppose that the batch is large or the production process is continuous, so that drawing a sample item with or without replacement has about the same result. In such cases, we can assume that the number of defective items x in a sample of sufficiently large size n follows a binomial probability distribution.

Now, if the producer is willing to sacrifice a little so that the batch with fraction defective AQL is accepted with a probability of at least (1-a), where a is a small positive number, and the consumer is willing to sacrifice a little so that the batch with fraction defective RQL is accepted with a probability of at most b, where b is a small positive number and RQL > AQL, then the following two inequalities can be established.

From the above two inequalities, the numbers n and c can be solved (but the solution may not be unique). The number 'a' is called the producer's risk, and the number ' b' is called the consumer's risk. As a common practice in industries, the magnitude of a & b is usually set at some value from 0.01 to 0.1. Nomo-gram is available for obtaining solution(s) of the inequalities given above easily.

The sampling plan is called a single sampling plan because the decision to accept or reject the batch is made in a single stage after drawing a single sample from the batch being evaluated.

Another type of sampling plan, which is different from the above, is called continuous sampling procedure (CSP). The rationale of CSP is that if we are unsure about the quality of products produced from a process, a 100% inspection will be adopted. If the quality of products is found to be good, only a fraction of the products will be inspected. In the simplest CSP, initially, 100% inspection is performed. During 100% inspection, if no defective items are found after a specified number of items are inspected (which means that the quality of the product produced is perhaps good), 100% inspection is stopped, and only a fraction f of the products is inspected. During fraction inspection, if a defective item is found (which means that the quality of products might have deteriorated), then fraction inspection is stopped, and 100% inspection is resumed. More refined CSPs have also been constructed, for example, by setting f at 1/2 at the first stage, 1/4 at the second stage, and so on.

Variables Sampling Plans

The sampling plans described above are called attribute sampling plans because the inspection procedure is based on a "go"/"no go" basis, meaning an item is either considered non-defective and accepted, or it is considered defective and rejected. Variable sampling plans are different from attribute sampling plans because continuous measurements, such as dimensional or weight checks, are made on each item in the sample. The decision to accept or reject the batch is then based on the sample mean or the average of the measurements obtained from all items contained in the sample. Variable sampling plans can be used, for instance, when a product item is deemed acceptable if a certain measurement x (e.g., diameter, length, hardness) exceeds a pre-set lower specification limit L; otherwise, the item is considered not acceptable.

7.8 SUMMARY

SPC aims at controlling the variability of process output using a device called the control chart. On a control chart, a certain characteristic of the product is plotted. Under normal conditions, these plotted points are expected to vary in a "usual way" on the chart. When abnormal points or patterns appear on the chart, it is a statistical indication that the process parameters or production conditions might have changed undesirably.

Process capability is the range over which the "natural variation "of a process occurs as determined by the system of common or random causes; that is, process capability indicates what the process can deliver under "stable "conditions when it is said to be under statistical control.

It is important to reiterate that acceptance sampling or sampling inspection is solely a means of screening to separate batches or lots of high quality products from those of poor quality. This screening process does provide some level of assurance of the quality of incoming parts and materials. The use of sampling plans enables industries to conduct this screening more efficiently than simply drawing samples at random.

As such, sampling inspection can be utilized during purchasing to check the quality of incoming materials when there is uncertainty regarding the conditions and quality control procedures in place at the vendor's plant. Acceptance sampling can also be employed for the final inspection of products after production. This process, to a limited extent, ensures the quality of the products before they are dispatched to the customer. However, it is important to note that unlike SPC, acceptance sampling does not assist in preventing the production of low-quality products.

7.9 KEY WORDS

Acceptance Sampling : The process of selecting a sample from incoming inputs (raw material, components etc.) or semi-finished, or finished products for the purpose of analysing the results.

Control Chart : A line chart showing the control limits at three standard deviations above and below the average quality level. It constitutes the core of SPC.

Design of Experiments : Application of statistical methods for producing high quality, robust products and process designs (one of the Taguchi methods).

Fishbone Chart : A diagram showing the main causes and sub-causes of a quality problem or defect.

Flow Chart : A device/method showing the order of activities in a process or a project and their interdependency.

Histogram : A Bar Chart showing the distribution of variable quantities or characteristics.

Pareto Chart : A chart showing the distribution of effects attributable to various causes or factors arranged from the most frequent to the least frequent.

Process Capability : The range over which the 'natural variation' of a process occurs and is determined by the system of common/random causes.

Statistical Process Control (SPC) : A process to control the variability of output using control charts.

Statistical Quality Control (SQC) : Use of statistical methods to improve or enhance quality for customer satisfaction.

7.10 SELF-ASSESSMENT QUESTIONS

1. Define statistical process control and discuss its advantages
2. What does the term statistical control mean? Explain the difference between capability and control.
3. What are the disadvantages of simply using histograms to study process capability?
4. Discuss the three primary applications of control charts.
5. Describe the difference between variables and attributes data. What types of control charts are used for each?
6. How should control charts be used by shop-floor personnel?
7. What are modified control limits? Under what conditions should they be used?
8. Describe some situtations in which a chart for indiviual measurments would be used.

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UNIT 8 TOOLS AND TECHNIQUES OF TQM

Objectives

After reading this unit, you should be able to:

- explain the need for benchmarking;
- implement and appreciate the usefulness of Quality Function Deployment (QFD);
- understand the concepts and tools that can be used to improve quality;
- appreciate the concept of reliability as a special dimension;
- understand the role of zero defects in Total Quality Management;
- understand the principles and practices of Re-engineering.

Structure

8.1 Introduction

8.2 Principles of Benchmarking

8.3 Types of Benchmarking

8.4 Quality Function Deployment (QFD)

8.5 House of quality (HOQ)

8.6 Reliability

8.7 5 'S'

8.8 Zero Defects (ZD)

8.9 Re-engineering

8.10 Taguchi Methods

8.11 Six Sigma Principle

8.12 Cross Functional Management

8.13 Hypothesis

8.14 Summary

8.15 Keywords

8.16 Self-Assessment Questions

8.17 Further Readings

8.1 INTRODUCTION

TQM employs several tools and techniques. We have discussed few of them. In this unit we will discuss Benchmarking and, Quality Function Deployment (QFD), Zero Defects, Reliability, Re-engineering, etc.

We now know that Total Quality Management (TQM) is essentially concerned with continuous improvement of the quality of goods and services to satisfy growing expectations of customers. Such improvements in final deliverables could only be effectively made when improvements are made in both design and delivery of deliverables. Various organizational variables like social, culture, human attitude, technology, etc. are also required to be suitably changed to effect improvements in the desired direction. This unit looks at some of these aspects. Reliability which is an important component of quality needs to be assessed, specified and improved with the help of suitable tools and techniques of reliability engineering. Reliability, therefore, is a major concern at the design stage. Similarly, appropriate steps are also required to be taken to improve design specifications, so that products could work satisfactorily even under adverse conditions, at least to a certain extent. Off-line Quality Control by the Taguchi Method aims at developing such robust designs. In spite of having a good design, a product may not ultimately satisfy customers unless the same is manufactured with sufficient care on the shop floor. Those working on the shop floor should inculcate good working habits so that they keep the place clean, properly organized and make efforts to see that the workplace continues to be pleasant to work at. A Japanese concept called 5 'S 'has drawn attention to this aspect. Similarly, management should also be sincere and should continuously facilitate all their employees to reach a level of perfection. Programmes like ' zero defects' are required to be taken by the management on a continuous basis. However, such improvements need to be supplemented sometimes by major initiatives which could cause improvements by an order-of-magnitude. Such a kind of initiative is known as 'Re-engineering'. We shall discuss all the above concepts and techniques in the following sections.

8.2 PRINCIPLES OF BENCHMARKING

The term Benchmarking has different meanings for different persons. To define benchmarking, it is "a standard against which something can be measured, i.e. a "reference point'. The term, as originally coined by Xerox Corporation, U.S.A., meant continuous process of measuring the products, services, and practices against the toughest competitors or organizations, known as 'leaders'.

This definition emphasizes the analytical part of the work measurement and comparison. But one of the subsidiaries of Xerox - Rank Xerox, U.K. looked at it somewhat differently. It viewed it as a continuous systematic process of evaluating organizations recognized as industry leaders, for determining business and work processes that represented best practices. In operational terms, this is frequently condensed to the search for industry best practices that lead to superior performance''. 'Best practices' means the methods and procedures used in work processes that best meet customer requirements. Benchmarking is not simply about what we want to achieve - the benchmarks or the measurements of best performance, but also how they are achieved i.e. processes in action. Thus, benchmarking is an ongoing task at all levels of business of finding and implementing the world's best practices in the key things that are done to deliver customer satisfaction.

The benchmarking process consists of the following three elements or components:

i) **Analysis:** breaking down an issue into components

ii) **Comparison:** component-by-component comparison to find gaps or differences.

iii) **Synthesis:** Recombination of components with ideas stimulated from the differences.

The original concept of benchmarking had been in practice in India about 3000 years ago. The dominant method of learning which was known then to people, used to be watching the actions of leaders. Foot prints of wise men are what common people are told to follow. Such a principle of learning got subsequently spread to other countries like Japan, and transformed the principle into a practical application in an industrial context. Japan calls it 'Dantatsu', a method of choosing the best of the best. Subsequently, US organization, faced with severe competition from Japanese organizations, made use of this principle to develop a structured method of adaptive innovation and termed it as 'Benchmarking'.

8.3 TYPES OF BENCHMARKING

Since benchmarking is essentially learning from comparison with others, it is categorized into different types depending upon either objects being compared or in respect of units with which comparison is made.

Based on the objects to be benchmarked, benchmarking has been classified into four categories. These categories are as follows:

- Product Benchmarking
- Performance Benchmarking
- Process Benchmarking
- Strategic Benchmarking

Product Benchmarking: This is also alternatively termed as 'Customer Satisfaction Benchmarking' or 'Customer Value Profiling'. This refers to both engineering and qualitative comparison of products and services among competing offerings. Many of the manufacturing organizations have been doing several kinds of reverse engineering to benchmarking various features of their products. Recently, several organizations especially in the service sector have shown their concern in comparing their services with others. Product Benchmarking can also help in identifying activities where improvement is possible. Product Benchmarking quite often leads to the redesign of existing products and services.

Performance Benchmarking: Performance benchmarking refers to comparison of performance indicators related to a business as a whole or a group of critical activities or processes. Business level performance benchmarking which was previously used under the banner of 'inter-organization comparison' done through a set of well defined financial ratios, has again drawn the attention of world class organizations. It has been found to be a very important tool to identify different functional areas where scope for improvement is high. However, such business level comparison now includes many performances issues like innovativeness of the organization, customer satisfaction from the product and services delivered by the organizations, etc.

Process level performance benchmarking includes comparison of process performance indicators in terms of those issues which are critical to the implementation of business strategy. These indicators relate to resource efficiency, resource allocations and many other issues which are relevant for the processes under reference. Performance benchmarking is becoming very important tool to provide external feedback to the concerned persons involved in the process or in any of the constituent activities.

Though improvement is implicit in performance benchmarking of both kinds mentioned above, there is no explicit attempt to design any new course of action as part of performance benchmarking.

Process Benchmarking: This refers to comparison of processes. A process is a set of sequential activities as performed on an item to increase its value to the customers. There are three major categories of processes - business process, support process and management processes, depending upon the nature of customers served. In a developed country where competition in product market has become fierce, business process benchmarking is fast becoming a standard management tool to maintain market share.

Process benchmarking requires explicit conduct of all three elements of benchmarking as mentioned above. In other words, it leads to the redesign of new processes to be implemented by the management.

Strategic Benchmarking: This refers to a study of corporate level or business level strategies pursued by some of the selected organizations reputed for reformulation of strategies or polices in recent times. Many of the top managements have been constantly reviewing the business portfolios of their organizations.

Benchmarking corporate strategies of other successful organizations could give some additional insights in their efforts. Business strategies refer to the direction in which a specific business is moving to ensure its attractiveness to customers and other stockholders. An organization could study the business strategy of another successful business and use the strategy for building up the attractions. Such strategic benchmarking will gain more and more significance as markets become competitive.

Based on the organizations against whom one is benchmarking, benchmarking methodology could again be categorized into five following types :

i) **Internal Benchmarking** - Comparison between department, plants, subsidiaries etc. within the country or among the countries through collaborations.

ii) **Industry Benchmarking** - Comparison of performances by the organizations producing the same class of products and services.

iii) **Competitive Benchmarking** - Comparison of performance against direct competitors.

iv) **Best-in-class Benchmarking** - Comparison with best practices prevalent in an organization irrespective of products and services.

v) **Relationship Benchmarking** - Comparison is made with an organization with which the benchmarking organization already has a relationship like customer-supplier relations, joint venture arrangement, etc.

Product Benchmarking

This refers to comparison of different features and attributes of competing products and services either through engineering analysis or through analyses of perception of customers. Engineering approach of product benchmarking was in practice for quite some time. This method, normally known as reverse engineering, is basically technical, engineering based approach comprising of tear-down and evaluation of technical product characteristics. Most of the consumer goods and capital equipment manufacturing organizations have one or other way, been doing reverse engineering to finalize product specifications. Some of the Indian organizations also carried out value analysis in conjunction with reverse engineering. Value analysis facilitates searching for cost-effective alternatives for the chosen components or sub-assemblies, specifications of which are found out through comparison process. An engineering department equipped with tools of value engineering, can develop suitable for developing product specifications that are comparable, if not better, to competitive offerings.

Comparison of customer-perceived-quality assumed great significance during the 1970's when it was revealed to be the most important factor for market share in competitive economic conditions. It was observed that customers made their judgment relative to value i.e. quality and price. The relative perceived quality is a quality from the customer perspective, with respect to alternative competitive offerings in the market.

Products and services provided by some organizations are of much lower perceived quality compared to world class offerings. Under this situation, one can easily perceive use of product benchmarking to identify immediate priority areas for improvement. This should be done on continuous basis so that all the important attributes as sought by customers are supplied through their offerings, thereby increasing relative customer-perceived-quality and, in turn, increasing market share.

Though some marketing departments of Indian organizations have been doing some form of product perception analysis, many of such analyses fall short of requirements as mentioned above. Further, it has been found that product benchmarking should be done by an inter-disciplinary group instead of marketing departments alone, to provide maximum benefits.

Performance Benchmarking

It is alternatively known as 'Result Benchmarking' or 'Benchmark Studies' as benchmarking often includes/ business or process outcomes. It is also termed as 'industry benchmarking' as participants of such a study often belong to a single industry.

For this kind of benchmarking, all the different kinds of system performance variables - efficiency, effectiveness, productivity, quality of work life, flexibility, innovativeness and profitability could be measured and compared either for a whole business or for any of the constituent functions or processes. Criticality in a business context decides which variables out of all the above mentioned ones are relevant and of common interest to all the participating organizations in a performance benchmarking study.

Performance benchmarking which was used to be known earlier as 'Interorganization Comparison' has been in practice in India since 1960s. In some of the industries like Textile, Fertilizer etc., performance benchmark's were in vogue. However, such comparisons included mostly financial ratios at aggregate levels and were often used for management control purposes. Performance benchmarking which has recently been prompted by the National Productivity Council of India includes all the strategic and critical issues both at business and funded levels that are current concern of managements of participating organizations. These issues are expressed either in financial or in operational terms.

Process Benchmarking

It is now realised that it is not enough to know 'where' an organization is but it is also important to know 'how 'and 'why' it has reached that stage. Process benchmarking provides us with a more effective and efficient process to be implemented. A process is a set of sequential activities performed on an item to add value to the item for creating customer satisfaction. Process benchmarking is a complete method containing all the elements of ideal benchmarking process i.e. analysis, comparison and synthesis. It is, thus, an essential tool when an existing process is to be redesigned or re-engineered.

There are many models available in the West regarding the appropriate methodology to carry out process benchmarking. National Productivity Council (NPC) of India has evolved a 8 step process to suit Indian conditions. According to the NPC mode1, process benchmarking methodology, to be carried out by a process oriented benchmarking team, consists of the following steps:

Step 1 : Identify the object or process to be redesigned or improved.

Step 2 : Map and measure the existing process in its entirety in terms of relevant critical dimensions.

Step 3 : Identify the partner where the same process is known to be better performed.

Step 4 : Analyse the partner's process and find out the differences. This often requires collection of data through a check list/questionnaire and/or physical visit to the partner's site.

Step 5 : Redesign the process and put up the proposal for management approval.

Step 6 : Implement the re-designed process .

Step 7 : Monitor the performance of the redesigned process.

Step 8 : Recalibrate the process.

Though the benchmarking partner could belong to the same organization or to the same industry, any major change in the process performance could be achieved when the partner is drawn from an organization whose products and services are quite dissimilar. However, in such benchmarking, acceptance by the organization personnel could be an important factor.

Several studies have shown that almost all the leading American and European organizations have gone for process benchmarking. In India and every process provides scope as each has been established based on traditional management thinking. An organization normally has three kinds of processes; business processes, support processes and management processes. The business process linking the organization, directly with customers is a high priority area for benchmarking in a highly competitive situation. But, quite often, an improved business process like distribution, customer service provision, etc. cannot be sustained over a long time if supportive processes are not equally efficient. So, support processes like procurement, manufacturing, machinery maintenance, etc. are required to be improved or re-engineered before an efficient business process is fully functional to its optimum performance level. In Indian circumstances, support processes should be priority areas for improvement. These also include management processes.

Strategic Benchmarking

Strategy is the ability to see where one wants to go, and to do things necessary to stay on track and get there.

Using this definition, strategy is both forward-looking (proactive) and side-looking (reactive). Looking around and learning from others are thus important enablers for such strategy planning.

Strategic thinking in business is the matching of a business's opportunities with its resources in order to develop a direction. Such thinking is often reflected in a strategic plan that specifies goals and objectives and areas for resource allocation. Strategic benchmarking addresses all the pertinent issues involved in the abovementioned strategic planning.

It is only during the 1990's that many Indian organizations have been seriously pursuing strategic planning and carrying out competitor's analysis. An established quantitative method of strategic benchmarking has been developed with the application of PIMS (Profit Impact of Market Strategy) model containing up-to-date strategic information of 3000 business units.

Alternatively, strategic benchmarking could also be carried out by the application of process benchmarking techniques and methods. Many strategic consultants in the West have now observed that strategic benchmarking is similar in application to benchmarking of work processes, but different in scope.

8.4 QUALITY FUNCTION DEPLOYMENT (QFD)

The level of competition is intensifying as the business environment shifts from national to global. This increase in competition has often focused on developing new and improved products to meet customers' specific needs. This is forcing organizations to match or surpass their competitors' products in terms of quality, price and service in order to survive. As customers become more value conscious and selective in the products they buy, organizations have begun to focus their attention on ways in which they could attain long-term market share and profitability through enhanced product development practices.

Examining how successful organizations develop their products may helpful to managers in understanding adopting benchmarking of their own product development methods in their organizations. Quality

Function Deployment (QFD) has been heralded as an important tool of the product development process.QFD enables an organization to measure customer 'wants' and map them against the engineering 'how' in a way that highlights trade-offs and drives the product's design toward customer requirements.

The importance of meeting customers' quality requirements has steadily migrated upstream through the manufacturing process since the final inspection era of the 1920s. As the cost of producing nonconforming products has risen (in terms of production cost, warranty claims, lost sales, etc.), organizations have attempted to identify the root causes. Statistical Process Controls (SPC), introduced in the army forces during World War II, and was one of the first methods used to reduce the number of defective products. SPC gained industry acceptance during the last 1950s and early 1960s and remained the most common form of quality control in the United States until the 1970s. While SPC can reduce the number of defective products that reach the customer, it is still a reactive quality control measure.

Designing quality into the product through improvements in the product development process and building robustness into the production system by designing products for easy manufacturing and assembly became important issues in the US during the 1980s. Addressing quality, production, and customer issues while still in the design stage of product development allows organizations to realize greater savings while attaining higher product quality. As the intensity of competition increases, organizations use cross functional teams with representatives from marketing, manufacturing, suppliers and final customers to design, market and manufacture products. This front-loading of quality allows organizations to develop and maintain a competitive advantage over competitors who follow the traditional 'over-the-wall' approach to product development.

QFD, which was first used at Mitsubishi's Kobe shipyard site in 1972, was designed as a tool to facilitate a cross-functional product development process. The application of QFD by Japanese organizations accelerated through the 1980s. The acclaimed benefits of using QFD (lower production cost, shorter development time, and higher quality) attracted a few US organizations in the mid to late 1980s. This gap between Japanese and US implementation of QFD methods may help to explain the superiority of Japanese products and the growth of Japanese organizations' market share during this time period.

What is QFD?

Quality Function Deployment has been defined as a system for translating consumer requirements into appropriate organization requirements at each stage from research and product development to engineering and manufacturing to marketing/sales and distribution (American Supplier Institute, 1989). QFD is taking the voice of the customer from the beginning of product development and deploying it throughout the organization through a sequence of phases. In QFD, the voice of customer aligns the organization's resources to focus on maximizing customer satisfaction and minimising waste. QFD is not just a quality tool, it is also a planning tool for developing new products and improving existing products.

QFD permits the 'voice of the customer', rather than the 'demands of management', to allocate organization resources and to coordinate skills and functions in producing the final product. Listening to the customer requires that the management works to gain an understanding of its customers at three levels. The first level understands the basic wants and needs of the customer. A cross-functional team using a variety of market research methods (e.g. individual interviews, focus groups and mail and telephone surveys) generates a list of customer requirements. The information collected during this stage is referred to as the 'spoken' quality demands and performance expectations. The requirements are usually vague (e..g. 'good ride') or incomplete and must be further defined into measurable characteristics. There is also information which is not directly given by the customer but must be included in the analysis to obtain a more complete understanding of customer requirements. The 'unspoken' attributes are often overlooked by the customer or assumed to be incorporated into the product (e.g. airplane arrives safely). The product must fulfill all these basic requirements, along with attaining high levels of performance in order to achieve a competitive level of customer satisfaction.

Second, QFD must drive the organization to go beyond these data collection techniques and identify fundamental customer needs and root product function. In addition to the spoken and implied wants of the customer, QFD forces the design team to determine hidden customer requirements by studying how customers use a product, examining the product's applications and learning customer behaviours. For example, customers may express a desire to have the bank offer more convenient hours. One response would be to open the bank for longer hours. Another would be to offer access to its services via a computer network and automated teller machines.

Third, in addition to learning the basic quality and performance attributes and root product functions, the product must provide unexpected features which 'excite and delight' the customer in order to sustain long term market share. These features are not usually known by the customer because they are either not aware of technological advances (e.g. new laser applications) or have become accustomed to standard product uses or applications. While some new characteristics have evolved from technological breakthroughs, not all new features must come through research and development. 'New' features or product applications are found when time is spent understanding the customer and how the product is being used. As new features are introduced and competitors copy or surpass existing products or services, characteristics which one surprised the customer, become standard product features over time (e.g. air bags in automobiles).

Hence, the list of product characteristics must be continually updated and revised.

8.5 HOUSE OF QUALITY (HOQ)

Figure 8.1 shows the essential components of the quality table or HOQ diagram. The construction begins with the customer requirements, which are determined through the 'voice of the customer' - the marketing and market research activities. These are entered into the blocks to the left of the central relationship activities. These are entered into the blocks to the left of the central relationship matrix. Understanding and prioritizing the customer requirements by the QFD team require the use of competitive and compliant analysis, focus groups, and the analysis of market potential. The prime or broad requirements should lead to the detailed WHATs.

Once the customer requirements have been determined and entered into the table, the important of each is rated, and the rankings are added. The use of the 'emphasis technique' or paired comparison may be helpful here.

Each customer's requirements should then be examined as per customer rating; a group of customers may be asked how they perceive the performance of the organization's product or service versus those of competitors. These results are placed to the right of the central matrix. Hence the customer requirements, importance rankings and competition ratings appear from left to right across the house.

The WHAT'S must now be converted into the HOWs. These are called the technical design requirements interactions.

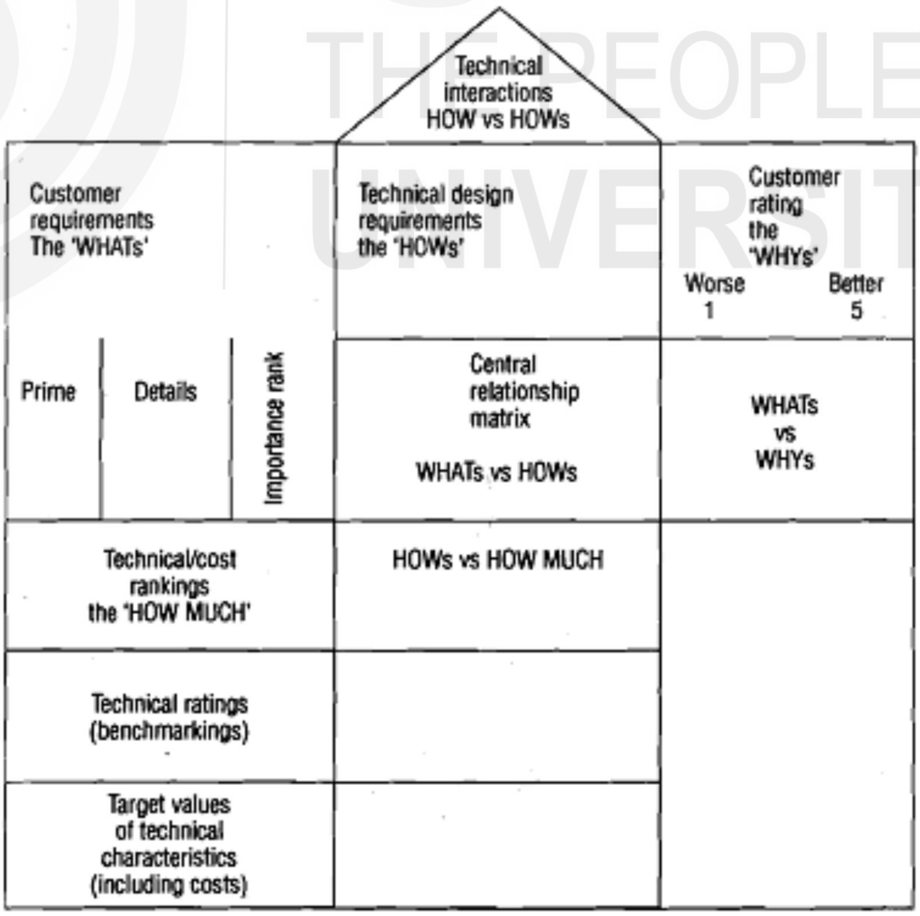

Figure 8.1: The House of Quality Table

They appear on the diagram from top to bottom in terms of requirements, ranking (or costs) and ratings against competition. These will provide the 'voice of the processes.

The technical requirements themselves are placed immediately above the central matrix and may also be given a hierarchy of prime and detailed requirements. Immediately below the central relationship matrix appear the rankings of technical difficulty, development time, or costs. These will enable the QFD team to discuss the efficiency of the various technical solutions. Below the technical rankings on the diagram comes the benchmarking data, which compares the technical process of the organization against its competitors.

The central relationship matrix is the working core of the house of quality diagram. Here the WHATs are matched with the HOWs, and each customer's requirement is systematically assessed against each technical design requirement. The nature of any relationship-strong positive, neutral, negative, strong negative – is known by symbols in the matrix. The QFD team carries out the relationship estimation, using experience and judgment, the aim is to identify HOW so that WHAT may be achieved. All the HOWs listed must be necessary and together sufficient to achieve the WHATs. The root of the house shows the interaction between the technical design requirements. Each characteristic matched against the others, and the diagram format allows the nature of relationships to be displayed. The symbols used are the same as those in the central matrix.

The complete QFD process is time-consuming, because each cell in the central and roof matrices must be examined by the whole learns. The team must examine the matrix to determine which technical requirement will need design attention, and the costs of that attention will be given in the bottom row. If certain technical costs become a major issue, the priorities may then be changed. It will be clear from the central matrix, if there is more than one way to achieve a particular customer requirement, and the roof.

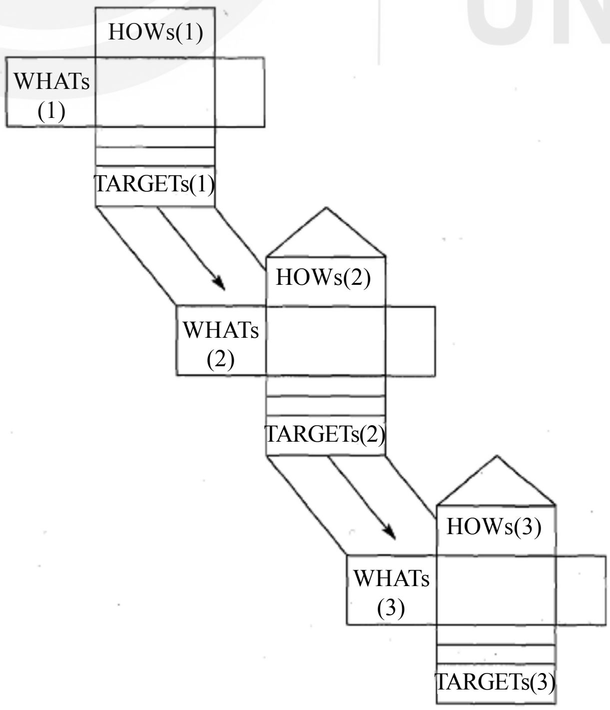

Figure 8.2: The 'Deployment' of the 'Voice of Customer' table

matrix will show if the technical requirements to achieve one customer requirement will have a negative effect on another technical issue.

The very bottom of the house of quality diagram shows the target values of the technical characteristics, which are expressed in physical terms. They can only be decided by the team after discussion of the complete house contents. While these targets are the physical output of the QFD exercise, the whole process of information-gathering, structuring, and ranking generates a tremendous improvement in the team's cross-functional understanding of the product/service design delivery system. The target technical characteristics may be used to generate the next level house of quality diagram, where they become the WHAT's, and the QFD process determines the further details of HOW they are to be achieved. In this way the process 'deploys' the customer requirements all the way to the final operational stages. Figure 8.2 show the target technical characteristics at each level become the inputs to the next level matrix.

An Abridged Example

A lawnmower manufacturer recently spent time and money redesigning a control for their most popular mower, only to find that customers were insensitive to the modification. So when they began planning to redesign another of their models, they wanted to be sure that this time round they would make changes the customer actually wanted.

1. Product planning matrix

- strong relationship = 3 points
- A medium relationship = 2 points
- weak relationship = 1 points

	How					
What	Motor	Drive Chain	Blades	Handle	Controls	Grass Collection
Quiet operation	●	●	△			
Works with wet grass	●	△	●			△
Cuts long grass	●	△	●			
Reliable	●	●			△	○
Light to carry	○	○	○	△		○
Compact for storage				●		●
Safe			●	△	●	○
Total	**13**	**11**	**12**	**7**	**5**	**8**

Figure 8.3: QFD for redesigning a lawnmower

The organization carried out market research to find out their customer's priorities, and a QFD analysis to see what the implications for design and manufacturing were. Figure 8.3 gives the details of QFD.

The results showed that customers were interested in performance and that improving the motor, the drive chain and the efficiency of the blades would have far more impact than changing the control features which is what was done.

There should be reductions in the number of midstream design changes, and these reductions in turn will limit post-introduction problems and reduce implementation time. Because QFD is consensus based, it promotes teamwork and creates communications at functional interfaces, while also identifying required actions. It should lead to-a 'global view' of the development process, from a consideration of all the details.

IF QFD is introduced systematically, it should add structure to the information, generates a framework for sensitivity analysis, and provide documentation, which must be 'living' and adaptable to change. In order to understand the full impact of QFD it is necessary to examine the changes that take place in the team and the organization during the design and development process. The main benefit of QFD is of course the increase in customer satisfaction, which may be measured in terms of, for example, reductions in warranty claims.

Activity 1

Is QFD or its some variant being used in organization of your choice? Discuss the format, if any, being used. In what ways has the concept of QFD benefitted the organization?

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8.6 RELIABILITY

Quality is a generic term meaning in conformance to requirements. When applied to a hardware product, it embraces all of the features and characteristics of a product that would satisfy customers. As goods and services are often used over a period of time, these features and characteristics need to manifest themselves rightly and consistently over a time. The time dimension of conformance is an important aspect of quality and this attribute is termed Reliability. The question of reliability becomes relevant if a product fails to conform to requirements after some period of initial conformance. Lack of reliability reflects malfunctioning of a product or failing of a product to perform within a specified period of time. In a fundamental way, the failures are natural in the whole universe as expressed in the second law of Thermodynamics. Sooner or later everything decays from an ordered state to a disordered state. Reliability is one of the important dimensions that have not been given as much emphasis in the west as in Japan he says.

Uncertain performance is reflected in hesitation in buying many Indian goods especially those manufactured by local small-scale industries, the reason being lack of reliability of the products.

Definition and Quantification

Since ‘reliability’ denotes the chances that a product will perform over time, its definition and quantification rely heavily upon the probability theory and statistics. The term reliability is defined as ‘the probability of a product/entity performing its specified functions under prescribed conditions without failure for a specified period of time’.

An analysis of this definition leads to the following prerequisites for the quantification of reliability:

1) The quantification of reliability must be done in terms of probability.

2) The definition must specify the environment in which the product must operate and for which reliability is quantified.

3) The definition must also include a statement of satisfactory product performances, that is, conformance to requirements.

4) The definition must also specify the required operating time period between failures.

Consistent with the above definition, reliability is measured in many ways. Two important measures are

“Mean time between failures’ (MTBF) and Failure rate per unit time. MTBF is suitable for a complete product that can be repaired. The product is operated under specified conditions and after some time it fails to meet one of its specified characteristics - usually a functional characteristic. The product is repaired and operation is restarted. After a further period of time it fails again. The procedure is continued and the times between failures are measured. The average of those intervals is MTBF.

Failure rate is a per unit measure for a mass-produced non-repairable product, such as electronic components.

To determine the failure rate of a population of components, say in a sample of 1000 components, a product is operated under specified conditions, and specified functional characteristics are tested either continuously or at predetermined time intervals. The accumulated number of failures is graphed against the product of the number of components on test and the time on test, (this product is being called “component hours”). The slope of this ‘survival’ curve at a particular time gives the instantaneous failure rate of the component sample at that time. The lower the failure rate, the better the reliability will be.

Customarily, MTBF is measured in hours and failure rate in reciprocal hours.

Reliability Engineering

Reliability is a design parameter just like weight, dimensions, or compressive strength, and it can be made a part of a requirement statement, translated into specifications, and submitted for verification by measurement and test. Therefore, the basic concern is to design the product with a specified reliability and to assure the customer that the product does pass the specified reliability. A discipline, called 'Reliability Engineering' has been evolved over the last fifty years to deal with these issues. This discipline essentially relates to failure statistics, testing, prediction and improvement in reliability.

Failure statistics as dominated in Reliability Engineering is mostly concerned with the concept of a probability distribution. Engineers find that different operating conditions and different products are better approximated by different mathematical forms. Among the most popular are the "exponential life function", which is alternatively, known as the "Bathtub curve". This is shown in the figure 8.4. The curve shows that an item may have higher possibility of failure in the beginning phases of its use as well as after its use over a considerable length of time. If the item is properly manufactured, it will have a little chance of failure during its most of the useful life

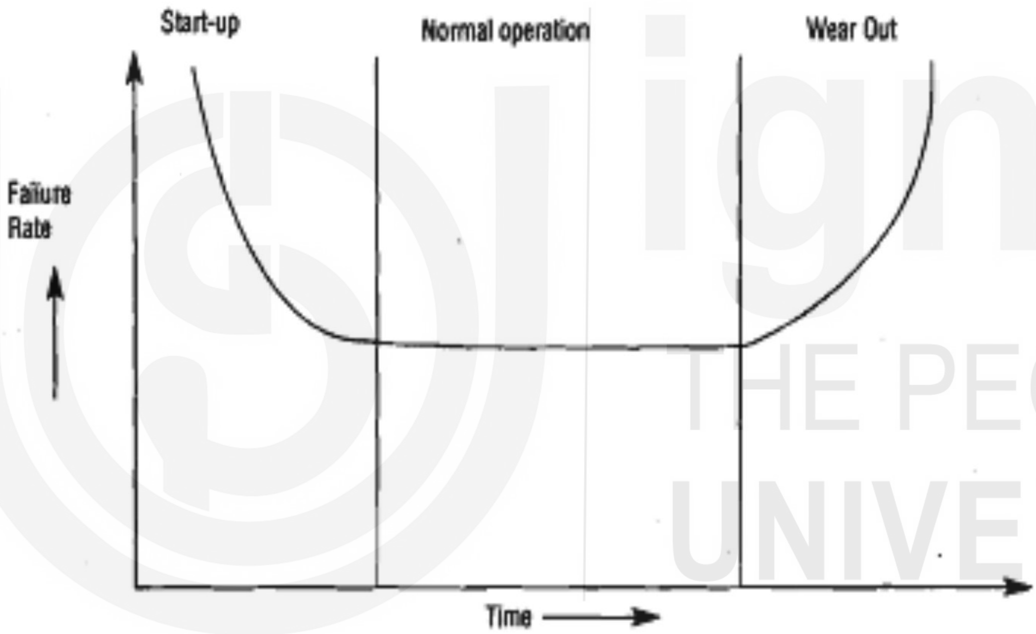

Figure 8.4: Reliability Curve

Prediction, however, is only the first step. The real goal is of reliability engineering to improve reliability and reduce failure rates over time. To accomplish these ends, a variety of techniques are employed: "Failure mode and effect analysis (FMEA)", which systematically reviews the ways a product could fail and on that basis proposes alternative designs; individual component analysis, which computes the probability of failure of key components and then tries to eliminate or strengthen the weakest links; dating, which " " requires that parts be used below their specified stress levels; and redundancy, which involves the use of a parallel system to ensure that backups are available whenever an important component or subsystem fails. An effective reliability programme also requires close monitoring of field. Failure. Otherwise, engineers would be denied vital information - a product's actual operating experience - useful for planning new designs. Field failure reporting normally involves comprehensive systems of data collection as well as efforts to ensure that failed parts are returned to the laboratory for further testing and analysis.

Application

Reliability normally becomes more important to consumers as downtime and maintenance become more expensive. It is therefore a very critical dimension of quality for consumer durable goods, capital equipment, and mission-critical items. Unreliable products not only drive away customers but also add cost.

Though Reliability Engineering as a serious practical science grew in US defence in 1950's, it was Japanese industries that paid serious attention to the subject subsequently. Japanese designers always try to reduce the number of parts per unit, following the basic principle of reliability engineering that, everything else being equal, the fewer the parts, the lower the failure rates. Parts are tested rigorously. 'Nasty tests" under varying conditions of temperature, humidity, and voltage are conducted, as with life tests, guided by actual field data showing the number of hours before a component typically fails. Changes in parts are often proposed as a result. All electric and electronic components sold in Japan are tested for reliability and the data are made available to others.

8.7 5'S'

5S occupies a prominent place as one of the basic tools to enhance the quality of the workplace. In fact, it forms the foundation for all improvement efforts as shown in the figure 8.5. It is an acronym for five Japanese words and denotes a step-by-step approach for developing a clean and well-organized workplace. The term workplace could mean shop floor, office, manager's desk, garage, store or any other place. It is more than housekeeping. 5S is very popular in Japan and is becoming common in other countries too. A well-organized workplace provides a healthy and safe work environment and also improves productivity by reducing time for locating tools and materials, damage to the materials, etc. A well-designed and implemented programme of 5S would instill discipline and change the attitude of employees toward work. It is considered one of the foundation level techniques for continuous improvement.

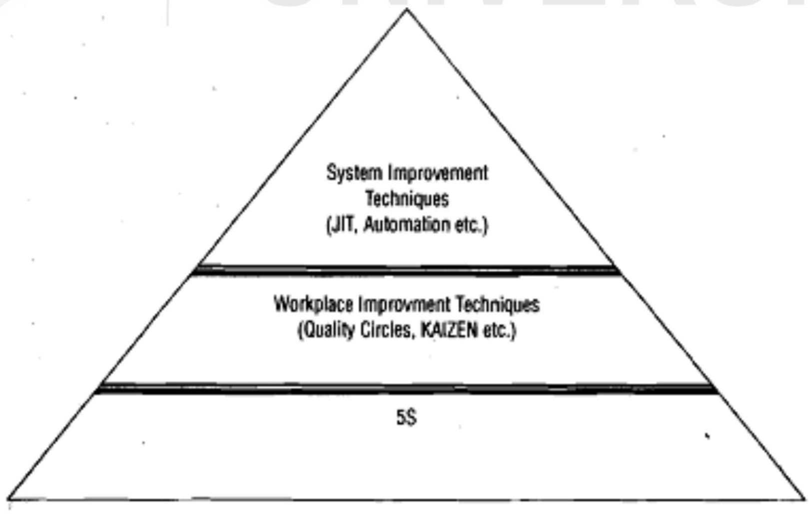

Figure 8.5: Quality Improvement Pyramid

The concepts that comprise 5S activities tend to be overly didactic. This is because the activities are not entered on results, but rather they emphasize people's behavioural patterns, such as the elimination of unnecessary items from the work environment or the cleaning and neatening of equipment. Consequently, the activities are of a kind that makes quantitative assessment of their effectiveness difficult. When 5S activities are promoted, questions are often raised such as "which is more important, 5SS activities or work?" or "How much earnings would 5S activities bring?"

5S activities are intended to qualitatively change the ways in which people think and behave, and, through these changes, to alter the quality of both equipment maintenance and the work environment. Before 5S activities are begun, everyone involved should clearly understand and codify in writing the goals and meaning of the 5S's for each individual organization and work environment. Furthermore, since 5S activities must be carried out with determination and concerted effort, it is helpful to devise slogans. Unless the specific rules are made highly pragmatic, it is hard to put them into practice. 5'S' represents five action items that ark initially done with some kinds of persuasion. However, these aspects should become a pan of the habit in individuals. These actions are described below:

1. SEIRI (Sort)

This denotes action to identify and sort out all items into necessary and unnecessary items and discard all unnecessary items from the shop floor. Items include parts, sub-assemblies, jigs, dies, raw materials, machines, files, documents in the files, racks, rejected goods lying in the workplace or "gemba" in Japanese.

Many supervisors and shop floor managers have a tendency to keep items which might be required in the distant future, or in greater quantities than necessary. The number of items required for a specific timeframe should be decided. Some of the steps comprise of

a) Classification of all items into categories - necessary (can be used in the near future) and unnecessary (not required at all or required in the distant future).

b) Fixing the quantities of items required in the distant future to be kept in the workplace.

c) Taking out trash and rubbish from the workplace immediately.

d) Prepare and circulate a list of items which are not required to all other departments/sections to identify what they can use.

e) Attaching a big red tag to all such items indicating name, disposal procedure, and date of disposal and retention period (the period after which items will not remain in the workplace).

f) Taking action to remove (from the workplace) dispose of these items after the retention period,

2. SEITON (Straighten)

This is the process of arranging all necessary items in an organized manner so that no time is lost in finding them. It involves deciding a place for everything and keeping everything in its place. The blowing guidelines should be helpful:

a) Depending on the frequency of their use, shelf life, bulk etc., decide how and where the item would be placed in the workplace. For example, small tools can be kept in racks while big bulky ones may be stored. More frequently used items should be kept in the middle shelves of the racks. The principle is to store them as near to the place of use as possible to minimize searching and handling time.

b) Use the first-in, first-out principle to prevent loss due to deterioration.

c) Put labels on all items to clearly identify them.

d) Store dies and moulds together along with tools necessary for setting them up.

e) Maintain a clear space around the fire extinguishers and keep passages free.

f) Clearly demarcate passages, work areas and storage areas.

g) Design storage methods so that items are stocked properly and are not damaged.

Establishing tidiness in the work environment depends upon a study of work efficiency. The amount of time needed to retrieve an item and return it to storage should be measured and the results can then be measured on a yardstick to evaluate the level of tidiness.

3. SEISO (Shine)

This is the process of cleaning the workplace thoroughly to remove all dust, oil, root etc. and waste from the machines, floors, walls, and other areas. Many times, cleaning leads to the discovery d potential Problem such are loose nuts, leakage and war. Implementation requires three steps. These are:

- Undertaking cleaning of the workplace completely with the participation of everybody including top and senior managers.
- Implementing schemes in your own machine and work by employees responsible for the designated areas.
- Ensuring that machines and workplaces are cleaned daily.

Cleaning is more than simply cleaning up of the general environment. It also means guarding from dirt every single part of the item used, upon which our livelihood depends. In other words, cleaning is a form of inspection. The problems caused by grime, dust, and foreign matter are related to all aspects of accidents, defects, and malfunctions. The recent tendency among young people to disdain dirty work environments is affecting personnel recruitment.

Cleaning should not be carried out by hired contractors. Rather, it should be an undertaking of the plant's employees, with responsibilities assigned to everyone. For large and public areas, a method of fair, rotational assignments is recommended. "Colour conditioning" i.e., choosing a pleasant paint scheme for the equipment and work environment, is an important element in creating a light and neat atmosphere. /

Personal cleanliness often refers to hygienic conditions for food and drink, such as the facilities of hot water and tea. Recently, an increasing number of shop floor workers have begun to wear light-coloured smocks. Since grime and dirt are readily seen on these smocks, the level of personal cleanliness can be easily judged. These cases exemplify the goal of personal cleanliness readily apparent.

In considering the environment and pollution in terms of the people affected, investigation of this subject expands into several other areas, all of which should be understood from the 5S perspective. These include dealing with oil, mist, dust, noise, paint-thinner-type oil products, and poisons. Some enterprises promote replacement of flowers in the shop environment. Thus, the idea of personal cleanliness embraces the upkeep of a generally clean and pleasant atmosphere.

4. SEIKETSU (Standardise)

This means maintaining a high standard of workplace organization and housekeeping at all times by repeating Seiri, Seiton and Seiso regularly. It involves the following:

a) Decide a schedule specifying frequency and persons responsible for carrying out Seiri, Seiton and Seiso and ensuring that it is done according to the schedule. Make 5S a routine.

b) Eliminate activities that make the workplace dirty and disorganized.

c) Protect workers from dangerous and unsafe conditions.

d) Standardize operations and daily maintenance procedures.

e) Institute competition between various sections departments for the best organized and maintained workplace - lay down criteria and determine the level of 5S activities in various workplaces.

f) Managers should take note of the status in their visits to the workplace.

iv) SHITSUKE (Sustain)

This denotes the self-discipline acquired by practicing 4Ss continuously, spontaneously and willingly to make them a part of daily life. It involves establishing standards, educating and training employees to observe good work habits and obey the rules of the workplace.

The three pillars of 5 ‘s’

Many think that 5 ‘S’ is a working philosophy. There are three pillars on which its implementation is dependent upon.

The first pillar to be focused upon is the disciplined work environment. The purpose of this goal is to focus on raising the basic management level through 5S practices. It can be seen as the essence of the 55’s themselves. 5S activities must transform the quality of human behavior. In a nutshell, the goal is to see that everyone abides by the rules once these rules are defined. Thus, of central importance are the ways in which everyone goes about carrying out his or her duties and participates in activities. Of equal importance are the training methods that allow each individual worker to be responsible for his or her assigned activities and behavior. Improvements clearly need to be made in the area of management by observation.

The key to the next step - making a clean work environment - is to clean the smallest elements of the work environment and equipment that have either never been cleaned or have never been thoroughly cleaned and to get rid of all the grime and dust. By thus radically transforming the condition of the work environment, the workers will be both motivated and inspired to renewed awareness and behaviour.

The third pillar of 5S is the creation of a work environment that can be managed by seeing, an idea popularly referred to as management by observation. By improving upon ways to identify abnormal conditions quickly and easily, it becomes possible to create an environment in which anyone in an area can spontaneously help someone else when an abnormal condition occurs. This means that measures are taken to prevent people from making mistakes and committing errors, and can be called the standardization of the 5Ss.

These activities that comprise the three pillars should be pursued simultaneously to take advantage of the mutual interrelationship among them.

Activity 2

Choose an organization which uses the concept of 5S. Prepare a table of the activities performed by the organization using 5S.

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8.8 ZERO DEFECTS (ZD)

There are numerous people engaged in a large number of activities related to marketing, design, manufacturing, distribution, etc. who contribute to make a product. If a product has to meet requirements of a customer, necessary instruction for every person involved requires description of every aspect of the product in precise, standardized and commonly understandable terms. These terms are known as specifications. Any variation that is caused while translating customers' requirements into right specifications or while producing needed specifications is considered as a defect. Unless special care is taken, such variations are normal.

Many people believe that it is uneconomical to eliminate all kinds of variations - major and minor ones -that are thought to normally happen at every work step. It is precisely the reason that acceptable quality level. (AQL) in statistical Quality Control has been used to define quality of goods and services in transactions.

A defect in missile could therefore be fixed, although it was likely to require extensive debugging before shipment. Subsequently, Martin's general manager accepted a request from the U.S. Army's missile command to deliver the missile one month ahead of schedule. He went even further - he promised that the missile would be perfect, with no hardware problems, no document errors, and all equipment set up and fully operational ten days after delivery (the norm was ninety days or more). Two months of feverish activity followed. Since little time was available for the usual inspection and after-the-fact correction of errors, all employees were asked to contribute to building the missile exactly right the first time. The result was still a surprise; a perfect missile was delivered. It arrived on time and was fully operational in less than twenty-four hours.

This experience was an eye-opener for Martin. After careful review management it was concluded that the project's success was primarily a reflection of its own changed attitude. The reason behind the lack of perfection was simply that perfection had not been expected. Once management demanded perfection, it happened! Similar reasoning suggested a need to focus on workers' motivation and awareness. Of the three most common causes of worker errors - lack of knowledge, lack of proper facilities, and lack of attention management concluded that the last had been least often addressed. It set out to design a programme whose overriding goal was to "promote a constant, conscious desire to do a job (any job) right the first time". The resulting programme was called zero defects.

Operating Principle

Crosby argues that if management is serious about achieving ZD, it has to be serious about prevention. He proposes some guidelines for managers which he calls the 'four absolutes of quality management'. These should be reflected in their attitude and management systems:

a) **Quality means conformance to the requirements:** The setting of requirements is management responsibility as are the communication devices and their effectiveness. Crosby argues that if management wants people to do things right the first time, they have to tell everyone clearly what that is; this must be expressed in objective terms as much as possible.

b) **Quality comes from prevention:** The first absolute was to understand the process by which various processes are involved in producing products/ services. The second is about identifying and eliminating all chances for an error to occur; this requires meticulous observation and analysis.

c) **Quality performance standard is Zero Defects:** This is conformance to the requirements and should be the personal performance standard of everyone in the organization according to Crosby, and will come from a change in attitudes; everybody needs to believe that it is always cheaper to do the job right the first time.

d) **Quality measurement is the price of non-conformance ie. cost of quality:** According to Crosby, manufacturing organizations spend 25% of sales doing things wrong and service organizations spend about 40% of their operating costs on the same wasteful actions.

Approach For ZD Implementation

To achieve great improvements, the management has to believe in the following:

- The conviction by senior managers that they have had enough of quality being a problem and wanting to turn it into an asset; pursuit of ZD will improve skill and knowledge of workers and managers.
- The commitment that they will understand and implement the four absolutes of quality management;

- The conversion to that way of thinking from the conventional wisdom that caused the problem in the first place.

Crosby argues that it takes a long time to transfer from conviction to conversion but that as soon as the transfer process begins, it is a positive sign that improvement has started to take place.

The Crosby approach to total quality is to change the culture and attitudes within an organization to implement continuous improvement. This approach is therefore more management - oriented than tool -oriented since it does not refer at all to the control of quality by the use of various statistical techniques. He has developed a management maturity grid based on management attitude to quality and the progression of quality performance achievements. An abridged version of above mentioned Quality Management Maturity

Grid is given in the Table 8.1. Crosby has suggested a 14- step quality improvement approach to reach the good of ZD. These steps are as follows:

i. **Management commitment:** Management should commit itself to participating in a quality improvement programme. Quality improvement never happens automatically but must be planned and managed.

ii. **Quality improvement team:** Representatives of each department should form a team. Its responsibilities embrace the whole quality improvement programme of the organization.

iii. **Quality measurement:** Determine the status of quality throughout the organization. Most quality improvement programmes depend on the systematic accumulation and analysis of quantitative data.

iv. **Cost of quality evaluation:** Establish the cost of quality to indicate where corrective action will be profitable for the organization.

v. **Quality awareness:** Share with employees the measurements of what non-quality is costing through training and communication material.

vi. **Corrective action:** Bring problems to light for all to see and resolve them on a regular basis where possible. It is for the supervisors to work out solutions.

vii. **Establish an ad hoc committee for the Zero Defects programme:** After a year has gone by, the observance of a Zero Defects Day will be organization management's commitment to 'Zero Defects' and also the thought that everyone should do things right the first time.

viii. **Supervisor training:** A formal orientation of the Zero Defects programme with all levels of management should be conducted prior to its implementation.

ix. **Zero Defects Day:** Zero Defects as the performance standard of the organization is observed on a day to provide emphasis and long lasting impression.

x. **Goal setting:** Regular meetings between supervisors and employees help people think in terms of meeting goal and accomplishing specific tasks as a team.

Table 8.1: Quality Management Maturity Grid

Issue Stage	Management understanding and attitude	Quality organisation status	Problem handling	Cost of quality as percent of salese
Stage I (Uncertainty)	No comprehension of quality as a management issue; lend to blame quality department for 'quality problems'	Quality is hidden in manufacturing or engineering departments; inspection probably not part of organization; emphasis on appraisal and sorting.	Problems are fought as they occur; no resolution; inadequate definitation; much yelling and accusations.	Reported: unknown Actual: 20%
Stage II (Awakening)	Recognizing that quality management may be of value but not willing to provide money or time to make it all happen.	A Stronger quality leader is appointed, but main emphasis is still on appraisal and moving the product; still part of manufacturing or other	Teams are set up to attack major problems; long range solution are not solicited	Reported: 3% Actual: 18%
Stage III (Enlightenment)	While going through quality improvement program, learn more about qality management; becoming suppotive and helpful	Quality department reports to top management, all appraisal is incorporated and manager has role in management of company.	Corrective action communication established; problems are faced openly and rresolved in an orderly way.	Reported: 8% Actual: 12%
Stage IV (Wisdom)	Participating; understand absolutes of quality management; recognize their personal role in continuing emphasis.	Quality manager is an officer of company; effective status reporting and preventive action; involved with consumer affairs and special assignments.	Problems are identified early in their development; all functions are open to suggestion and improvement.	Reported: 6.5% Actual: 8%
Stage V (Certainty)	Consider quality management an essential part of company system.	Quality manager on board of directors; prevention is main concern; quality is a thought leader	Experts in the most unusual cases, problems are prevented	Reported: 2.5% Actual: 2.5%

xi. **Removal of error causes:** Individuals are asked to describe any problems that keep them from performing error-free work. The appropriate functional group will develop an answer to those problems.

xii. **Recognition:** Award programmes are established to recognise those who meet their goals or perform outstanding acts. Awards need not be financial, recognition is what is important.

xiii. **Quality councils:** Quality professionals and team chairpersons should meet regularly to communicate and determine actions to upgrade and improve the quality improvement programme.

xiv. **Do it again:** Set up a new team of representatives and begin again to overcome the turnover and changing situations that can occur in the year to 18 months to implement the typical quality improvement programme.

The above-mentioned quality improvement philosophy has found its application all over the, world in many industries like Automobile, Medicine, Defense, etc. It requires sustained efforts on the part of management.

8.9 RE-ENGINEERING

Concept

It was only during 1980's western management got convinced to the effect that a process i.e. a planned set of actions effect product quality considerably as shown in Figure 9.5. A process converts inputs to outputs with the help of transforming resources. Unless proper resources, and inputs are properly acted upon, output quality leading to customer satisfaction could not be achieved. Such processes could be visualized at different levels - Business Process, Functional Process, Work Process, etc. depending upon the span of work. Business Process links up organizational activities to deliver specified set of goods and services to a customer. Hence, considerable amount of efforts are now made to improve upon such a business process in order to improve quality. Information Technology has been extensively used to modify a business process so as to deliver required services to customers. Often major modification is required to be made so as to ensure delivery of quality goods and services. Such major change to a business process is termed as Business Process Re-Engineering.

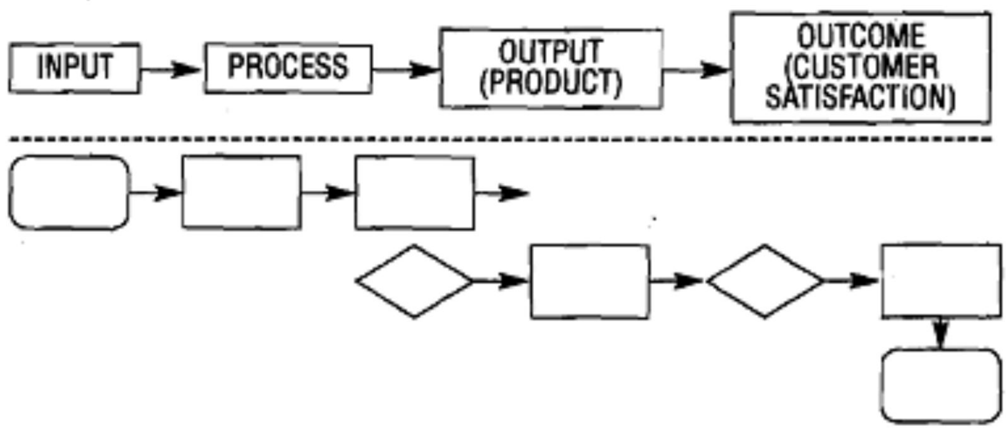

Figure 8.6: Process Concept for Quality

According to Hammer and Champy, Business Process Reengineering (BPR) is the fundamental rethinking and radical redesign of business processes to achieve dramatic improvements in critical and contemporary measures of performance, such as cost, quality, services and speed .

The above definition which is accepted by all could be elaborated further if the four key components of the definition are explained to a greater extent in the light of the understanding which western countries have built up over the years as given below:

a) **Fundamental rethinking and radical redesign signifies** an innovation that is a new way of thinking and acting. Fundamental rethinking refers to an enquiry on the most basic thing and asking fundamental questions about the existing methods and prevailing practices in organizations. Thus, one gets to the root of the things: not making superficial changes or refinements with what is already in place. People who are conversant with work study remember the purpose of asking a basic question -'WHY?' Finding an answer related to the fundamental reason for an action helps one make a breakthrough improvement.

b) **Business process** is a sequence of activities or operations performed on one or more inputs to deliver an output. It is a customer/demand driven, technologically ordered set of operations. For example, an order fulfillment process includes several tasks which are required to be carried out by different functional departments of an organization. Such cross-functional set of tasks is often the focus for BPR.

c) **Dramatic improvement** reflects achievements in terms of customer valued performance measures. It implies not to be content just with 10 or 20 per cent improvement in a certain activity or process but to get 10 times improvement and so on.

d) **Critical and contemporary measures of performance** highlight the criteria which dominant stakeholders in the prevailing markets look for. This includes an assessment of the bargaining strength of various stakeholders as well as the priorities in their expectations. This part of the definition makes BPR a dynamic and context-dependent contingency management approach. There are many other definitions provided by experts throughout the world. Most of these definitions differ in emphasis.

Characteristics of Reengineering

The following are the common features among BPR applications as found by Hammer and Champy .

Several jobs are combined into one

In this case one person will be doing all the tasks. Such a person will be known as a case worker. In some cases it is not possible to compress all the steps of a lengthy process into one integrated job performed by a single person. In those situations, the organization may require more persons, each managing parts of a process. Such a team will be known as a process team. In both the cases, the worker involved should be multi-skilled as his work becomes multidimensional.

Processes have multiple versions

This refers to the end of standardisation. To meet the demands of today's environment, we need multiple versions of the same process, each one tuned to the requirements of different market situations or inputs. Traditional one-sizefits-all processes are usually very complex, since they must incorporate special procedures and expectations to handle a wide range of situations. A multi-version process, by contrast, is clean and simple because each version needs to handle only the case for which it is appropriate.

Work is performed where it is directly needed

This means that work should be shifted across the organizational boundaries, i.e. the work should be done at the place where it is required most. Hammer cited the example of purchasing stationery items. When done by a central purchasing department, there is an inevitable delay. So, purchasing has to be done by the user department which requires the stationary items.

Non-value adding activities are reduced

For effective reengineering, checks and controls should be minimized. Reengineering processes use controls only to the extent they make economic sense. In the reengineering environment a larger number of people are empowered, so people become responsible for the end product.

Reengineering organization may look like a cent raked as well as decent raked system

Organizations which have reengineered their processes have the privilege to get the advantages of centralization and decentralization in the same process mostly on account of availability of modern IT facilities.

In fact, the above mentioned six characteristic from the basic design principles for BPR.

Methodology

There is a wide range of methodology through which an organization seeks to synergise some of the ideas implicit in the reengineering concept. While some methods focus on the analysis and redesign tasks, others focus on the definition of strategy or the development of the underling information system.

Given the wide variety of methods in practice, a typical one developed by Talwar (1994) is outlined below.

This methodology seeks to strike a balance between strategy formulation, process redesign, and exploitation and management of the reengineered business. It entails three stages below.

- **Programmeme Initiation:** Objectives and scope for the study are delineated.
- **Design and implementation:** Processes as decided under the scope are redesigned and prototypes are formulated.
- **Exploitation of the re-engineered business:** The redesigned processes and changes it enables are implemented.

Programme Initiation

- Build the awareness of business processes, create an understanding of the concepts of reengineering and, generate an interest in the need for, and opportunities to achieve sustainable performance improvement.

- Clarify the strategic direction and target architecture. Determine the strategic objectives for the exercise. Decide on the scale of change required and the number and scope of the candidate processes for reengineering. Define the business requirements and performance objectives that the re-engineered processes will fulfill and select a programmeme owner and project teams (s).
- Undertake the detailed planning of the change process.

In short, the first stage prepares the ground for undertaking a reengineering exercise.

Design & Implementation

- Map, model and analyse the current process (es), capturing performance measures to help highlight the need for change and determine if immediate improvement can be made. Identify the customer requirements and expectations from the process and (possibly) benchmark the process. In analyzing a process it is important to distinguish between the flows of activity and control in the process and the ways in which data and documents are used in the process.
- Generate a range of options for redesigning the process, each of which will significantly improve performance,typically using different configurations of process roles, work tasks and technology support.
- Test the options against key measures, such as customer requirement, cost and quality objectives. Select the preferred option and develop the business case. Develop the redesigned business process (es), develop or enhance the underlying computer systems add plan the migration steps.
- Prototype, test and refine the redesign in a laboratory setting; train the users and integrate the reengineered processes into the live business environment; test them in operation and implement any immediate refinements.

Thus, the stage undertakes all the technical aspects of reengineering.

Exploitation of the Reengineered Business

- Provide ongoing support to the reengineering operations, help managers and staff adjust to new roles, responsibilities and methods of working; monitor performance an’d initiate ongoing refinements.
- Ensure a continuous review of the opportunities to exploit the enhanced capabilities created through reengineering.
- Maintain a balance between continuous incremental improvements and periodic fundamental changes in process design.

The last stage is thus concerned with implementation and periodical review.

Application

A developing country like India offers vast scope for application of BPR as organizational capabilities in most sectors required for the market economy do not quite match the competitive demands. Most organizations which have hitherto been operating under a protected environment are predominantly bureaucratic in structure and style, and operating systems consist of age-old clerical operations carried out in sequential desk-to-desk manner. Rising competition is expected to force organizations to change all this.

Severe competition makes an organization customer-driven. A thorough study of the market and the strategic capability of an organization is therefore essential to pinpoint the areas requiring priority actions for changes.

BPR involves changes in organization structure, roles and responsibilities, rewards and recognition systems, etc. Such changes pose great challenges to Indian organizations. However, given the perceived threat and eagerness to change the age-old work practices, especially in the emergent liberalised economic situation, coupled with the entry of multinationals in many sectors, Indian management is believed to be willing to adopt BPR. With the rise in the educational level of future employees, and a variety of IT hardware -and software on offer, BPR may become a preferred choice for Indian management for effecting a strategic breakthrough.

Activity 3

Is your organization or any other organization you know of or familiar with have any of the five Concepts/approaches discussed in this unit viz., Reliability, Off-line quality control, 5S, Zero defects, and Reengineering. Have any of these concepts/ideas/approaches been applied? Discuss the state of the art, as it exists in the organization concerned, highlighting the salient features of each and the benefits the organization has derived from its application.

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8.10 TAGUCHI METHODS

In this section we discuss briefly the methods that belong to the domain of quality engineering, a recently formalized discipline that aims at developing products whose superior performance delights only when the package is opened, but also throughout their lifetime of use. The quality of such products is robust, i.e., it remains unaffected by the deleterious impact of environmental or other factors often beyond the users' control.

Since the topic of quality engineering is of notably broad appeal, we include below a brief review of the associated rationale and methods. The term "quality engineering" (QE) was used till recently by Japanese quality experts only. One such expert is Genichi Taguchi (1986) who reasoned that even the best available manufacturing technology was by itself no assurance that the final product would actually function in the hands of its user as desired. To achieve this Taguchi suggested the designer must "engineer" quality into the product, just as s/he specifies the product's physical dimensions to make the dimensions of the final product correct.

QE requires systematic experimentation with carefully developed prototypes whose performance is tested in actual field conditions. The object is to discover the optimum set-point values of the different design parameters, to ensure that the final product would perform as expected consistent in actual use. A quality-engineered product has robust performance.

Taguchi's Thoughts on Quality

Taguchi, a Japanese electrical engineer by training, is credited to have made several contributions in the management and assurance of quality. Taguchi studied the methods of design of experiments (DOE) at the Indian Statistical Institute in the 1950s and later applied these methods in a very creative manner to improve product and process design. His methods now form the foundation of engineering design methodology in many leading industries around the world, including AT&T, General Motors and IBM.

Taguchi's contributions may be classified under the following three headings:

- The loss function
- Robust design of products and production processes
- Simplified industrial statistical experiments

The essence of the loss function concept may be stated as follows. Whenever a product deviates from its target performance, it generates a loss to society. This loss is minimum when performance is right on target, but it grows gradually as one deviates from the target. Such a philosophy suggests that the traditional "if it is within specs, the product is good with a view of judging a product's quality is not correct".

If your foot size is 7, then a shoe of size different front 7 will cause you inconvenience, pain, loose fit, and even embarrassment. Under such conditions it is meaningless to seek a shoe that meets a spec given as (7 + x).

To state again, the loss function philosophy says that for a producer, the best strategy is to produce products as close to the target as possible, rather than aiming at "being within specifications."

The other contributions of Taguchi are a methodology to minimize performance or quality problems arising due to non-ideal operating or environmental conditions, and a simplified method known as orthogonal array experiments to help conduct multi-factor experiments toward seeking the best product or process design. These ideas may be described as follows.

The Secret of Creating a Robust Design

A practice common in traditional engineering design is sensitivity analysis. For instance, in traditional electronic circuit design, as well as the development of performance design equations, sensitivity analysis of the circuit developed remains a key step that the designer must complete before her/his job is over.

Sensitivity analysis evaluates the likely changes in the device's performance, usually due to element value tolerances or due to value changes with time and temperature.

Sensitivity analysis also determines the changes to be expected in the design's performance due to factor variations of uncontrollable character. If the design is found to be too sensitive, the designer projects the worst-case scenario to help plan for the unexpected. However, studies indicate that worst-case projections or conservative designs are often unnecessary and that a "robust design can greatly reduce off target performance caused by poorly controlled manufacturing conditions, temperature or humidity shifts, wider component tolerances used during fabrication, and also field abuse that might occur due to voltage frequency fluctuations, vibration, etc.

Robust design should not be confused with rugged or conservative design, which adds to unit cost by using heavier insulation or high reliability, high tolerance components. As an engineering methodology robust design seeks to reduce the sensitivity of the product process performance to the uncontrolled factors sthrough a careful selection of the values of the design parameters. One straightforward way to produce robust designs is to apply the "Taguchi method".

Robust Design by the "Two-step" Taguchi Method

Note that a product's performance is "fixed" primarily by its design, i.e.,by the settings selected for its various design factors. Performance may also be affected by noise/environmental factors, unit to unit variation in material, workmanship, methods, etc., or due to aging/deterioration. The breakthrough in product design that Taguchi achieved renders performance robust even in the presence of noise, without actually controlling the noise factors themselves. Taguchi's special "design 'noise array 'experiments discover those optimum settings. Briefly, the procedure first builds a special assortment of prototype designs -(as guided by the "design array") and then tests these prototypes for their robustness in "noisy" conditions. For this, each prototype is "shaken" by deliberately subjecting it to different levels of noise (selected from the "noise array", which simulates noise variation in field conditions).Thus performance is studied systematically under noise in order to find eventually a design that is insensitive to the influence of noise.

To guide the discovery the "optimum" design factor settings Taguchi suggested a two-step procedure. In Step 1, optimum settings for certain design factors (called "robustness seeking factors") are sought so as to ensure that the response (for the bar, plasticity) becomes robust (i.e., the bar does not collapse into a blob at least up to 50°C temperature). In Step 2, the optimum setting of some other design factor (called the adjustment" factor) is sought to put the design's average response at the desired target (e.g., for a plasticity level that is easily chewable).

Taguchi Orthogonal May Experiment

Rather typically, the performance of a product (or process) is affected by a multitude of factors. It is also well-known that over 213 of all product malfunctions may be traced to the design of the product. To the extent basic scientific knowledge allows the designer to guide his/her design, the designer does her/his to come up with selection of design parameter values that would ensure good performance.

Frequently though, not everything can be predicted by theory and experimentation or prototyping must be resorted to and the design must be empirically optimized. The planning of such experimental multiple factor investigations falls in the domain of statistical design of experiments (DOE).

Quality Control by Taguchi's Method

A typical quality system of a manufacturing organization consisting of three aspects - quality of design, quality of conformance, and quality of service - could be broken down into two parts as follows:

- Off-line Quality Control (QC): Activities for quality of design through market research and Product/process development, which are QC efforts away from production lines.
- On-line Quality Control (QC): Activities for quality of conformance and quality of service through manufacturing care, inspection and customer service. These are QC efforts mainly on production lines. Reliability Engineering as explained in an earlier section is one way of off-line quality control activity.

One of the well-known Japanese quality gurus, Genichi Taguchi, has drawn attention to another off-line quality control activity that aims at developing an optimum set of product and process conditions that would produce more uniform goods and services. According to him, any deviation from a target value of a critical dimension would cause a loss either to the producer or to consumer i.e. to society. He strongly recommends close uniformity among parts, components and products so as to cause minimum loss to a society. His major focus has therefore been on the design stage.

Principles and Methods

According to Taguchi, designing should consist of the following three activities

a) System design.

b) Parameter design

c) Tolerance design

The method starts up with system-design reasoning. This involves both product design and Product design.

Product design induces choice of materials and parts, and includes the calculating starting figures for product levels. Process design induces choice of equipment and taking a first shot at deriving process factor levels. Next, the method calls for parameter design. The aim is to find, from the start points already determined, an optimal mix of product parameter levels and process operating levels. Emphasis is placed on sensitivity to environmental perturbations and anything else that cannot normally be dealt with using control procedures of the organization. Reducing sensitivity is the aim. Securing minimum sensitivity relies on the help of techniques. Taguchi's techniques include ones that are efficient in searching for the optimal, workable, survival mix. Tolerance design then builds in tolerance to factors that can significantly affect the variations of the product. This might mean investment of money to improve equipment and materials. Tolerance design concludes with prototyping.

Once prototyping is complete, implementation is undertaken. At this stage use of Statistical Quality Control (SQC) would be relevant. Quality in design and process control clearly follow on.

'Robust design' uses the following statistical design optimization tools:

- Matrix Experiment
- Analysis of Mean (ANOM)
- Analysis of Variance (ANOVA)
- Signal-to-Noise (SIN) Ratios.

a) **Matrix Experiment:** It is the minimum set of experiments which are required to be performed with combinations of different levels of control factors (design parameters) so that it is possible through data analysis to predict the best combination of the control factors which make the variance due to noise factors a minimum. The matrix experiment is based on orthogonal arrays taken from linear algebra.

b) **Analysis of Mean (ANOM):** It is a data analysis method which helps to estimate the contribution of each level of every control factor to the Signal-to-Noise ratios. With the help of ANOM it is possible to predict the optimum combination of control factor levels to get quality-cost optimization.

c) **Analysis of variance (ANOVA):** It is a data analysis method which help to estimate the strength of each control factor relative to other in influencing variation in the Signal-Noise Ratio (variance).

d) **Signal-to-Noise (SIN) Ratio:** It is a new measure of quality as seen from the customer point of view.

The SIN Ratios were first proposed by Taguchi and form the basis for the improvement in quality purely through experiments. Maximising SIN ratio is equivalent to reducing variance due to various noise factors and hence improves quality during manufacturing, customer usage and aging and simultaneously reducing cost substantially.

Robust Design aims at finding, through structured experiments and data analysis, the combination of design parameters which give a maximum value of SIN ratios.

Strengths and Weaknesses of Taguchi's Method

The main strengths are:

- Taguchi's method pulls quality right back into the design stage rather than relying on adjustments to the process on-line, or inspection of the final product.
- The philosophy recognises quality as a societal issue and not just an organizational one. Quality is the minimum loss imparted by the product to society from the time the product is shipped.

- The method is developed for practicing engineers rather than theoretical statisticians.
- The method helps an organization to determine the best method of process control, reducing process costs incurring only small amounts of capital expenditure.
- The main weaknesses are:
  - ❖ The method has little relevance to processes where measurement does not produce data that can be accessed through sensitivity analysis and minimization; i-e., it is not relevant to many important success factors in the service sector.
  - ❖ There are no directives about how to manage a quality-oriented organization, particularly the people. Quality is placed organizationally in the hands of a specialised team of expert designers and does not involve managers and workers. Taguchi's method is designed for engineers and not managers.
  - ❖ Taguchi has nothing to say about humans as social, cultural or political beings and therefore makes no contribution to managing employees.

Inspite of many limitations, Taguchi method is finding increasing applications.

Applications

Every application of Robust Design has given incredible results. The following are the few examples of its application mostly in U.S.A which indicate the incredible power of this method for improving quality and reducing costs.

(ii) Window Photolithography

This was the first application in the USA that demonstrated the power of th-e Robust Design approach to quality cost optimization. The benefits achieved were

- 4 fold reduction in process variance
- 3 fold reduction in fatal defect
- 2 fold reduction in process time
- Easy transition - design to manufacturing:

(iii) Polysilicon Deposition Process

This process is one of the important steps in the manufacturer of very large scale integrated (VLSI) circuit. The existing process gives as high as 5000 defects per units' surface area of the silicon wafers. In all, 6 design parameters like furnace temperature, gas flow, gas pressure, settling time and cleaning method, were investigated with 18 experiments. The project was completed in 9 days. The benefits were

- Surface defect count was consistently reduced to less than 10 defects per unit surface area

- Yield improved manifold
- One expensive on-line inspection position was completely removed.

iv) Drilling a Hole in Cast Alloy Wheel

The existing drilling process gave heavy rejection of cast alloy wheels of racing cars due to high variance in the diameter of drilled hole in the wheel for installing the air valve stem. The benefits were

- diameter variation Gas reduced by 70%
- US 60000 dollars were saved.

There are a large number of applications in U.S., Europe & Japan, as above that could be illustrated. A few organizations in India, who have seriously tried to apply Robust Design, have obtained very encouraging results in their first trial applications. These organizations mostly belong to the Telecommunication sector.

8.11 SIX SIGMA PRINCIPLE

The six sigma principle is Motorola's own rendering of what is known in the quality literature as the zero facts (ZD) programme. Zero defects is a philosophical benchmark or standard of excellence in quality proposed by Philip Crosby. Crosby explained the mission and essence of ZD by the statement. What standard would you set on how many babies nurses are allowed to drop?" ZD is aimed at stimulating each employee to care about accuracy and completeness, to pay attention to detail, and to improve work habits. By adopting this mind-set, everyone assumes the responsibility toward reducing his or her own errors to zero.

One might think that having three-sigma quality, i.e., the natural variability equal to tolerance (= upper spec limit - lower spec limit, or in other words, Cp = 1.0) would mean good enough quality. After all, if distribution is normal, only 0.27% of the output would be expected to fall outside the product's specs or tolerance range. But what does this really mean? An average aircraft consists of 10,000 different parts. At 3-sigma quality, 27 of those parts in an assembled aircraft would be defective. At this performance level, there would be no electricity or water in Delhi for one full day each year. Even four sigma quality may not be OK. At four-sigma level there would be 62.10 minutes of telephone shut down every week.

You might wish to know where performance is today. Restaurant bill errors are near 3-sigma. Payroll processing errors are near 4-sigma. Wire transfer of funds in banks is near 4-sigma and so is baggage mishandling by airlines. The average Indian manufacturing industry is near the 3-sigma level. Airline flight fatality rates are at about 6.5 sigma level (0.25 per million landings). At two-sigma level an organization's cost of returns, scrap, rework and erosion of market share costs over a third of its yearly sales.

For the typical Indian, a 1-hour train delay, an incorrect eye operation or drug administration, or no electricity or water half a day is no surprise; helter routinely experiences even worse performance. Quantitatively, such performance is worse than two-sigma. Can this be called acceptable? One- or two sigma performance is downright noncompetitive.

Besides adopting TQM as the way to conduct business, many organizations worldwide are now seriously looking at six-sigma benchmarks to assess where they stand. Six sigma not only reduces defects and raises customer acceptability; it has been now shown that it can actually save money as well.

The Step to Six Sigma

Motorola prescribes six steps to achieve the six-sigma state, as follows.

Step 1 : Identify the product you create or service you provide.

Step 2 : Identify the customer(s) for your product or service and determine what they consider important.

Step 3 : identify your needs to provide the product or service that satisfies the customer.

Step 4 : Define the process for doing the work.

Step 5 : Mistake-proof the process and eliminate waste effort.

Step 6 : Ensure continuous improvement by measuring, analyzing and controlling the improved process.

Many organizations have adopted the Measure-Analyze-Improve-Control cycle to step into six sigma.

Typically they proceed as follows:

- Select critical-to-quality characteristics
- Define performance standards (the targets to be achieved)
- Validate measurement systems (to ensure that the data is reliable)
- Establish Product Capability (how good are you now?)
- Define performance objectives
- Identify sources of variation (seven tools etc.)
- Screen potential causes (correlation studies, etc.)
- Statistical Quality Control
- Discover relationship between variables causes or factors and the output (DOE)
- Establish operating tolerances for input factors and output variables
- Validate the measurement system
- Determine Process Capability (CPK) (Can you deliver? What do you need to improve?)
- Implement process controls

One must audit and review nonstop to ensure that one is moving along the charted path.

The aspect common between six sigma and ZD ("zero defects") is that both concepts require the maximum participation by-the entire organization. In other words, they require that unrelenting effort by management and the involvement of all employees. Organizations such as General Motors have used a four-phase approach to seek six sigma. These are:

1. **Measure:** Select critical quality characteristics though Pareto charts; determine the existing frequency of defects, define target performance standard, validate the measurements system and establish existing process capability.

2. **Analyze:** Understand when, where, and why defects occur by defining performance objectives and sources of variation.

3. **Improve:** Identify potential causes, discover cause-effect relationships, and establish operating tolerances.

4. **Control:** Maintain improvements by validating the measurement system, determining process capability, and implementing process control systems.

8.12 CROSS FUNCTIONAL MANAGEMENT

Hierarchical organizations are designed to manage the vertical flow of tasks. The horizontal flow between departments is managed through "co-ordination" efforts of managers who handle multiple departments and through meetings, committees or task forces. None of these structures has ever worked altogether satisfactorily, and therefore many attempts have been made to reform this structure.

The basic problems are two-fold. One, the processes by which work gets done in the organizations are horizontal, that is, cross-departmental. Almost anything worthwhile in an organization need to move through multiple departments or sections. Cost, Delivery, Safety and Morale (Q,C,D,S and M) require horizontal or cross-departmental management.

The first to see the real business organization as horizontal rather than vertical was Deming who in 1950, sketched out what he called the organization structure. Though the processes flow horizontally, the structure of an organization can turn into what has been described as functional, where departments can become fortresses defending their positions and thus interfering with the primary task of satisfying customers.

In Japan, the challenge of cross-functional management was first taken by Toyota with the help of Ishikawa. The result was what is called in TQM as cross-functional management. For example, Toyota viewed the functions of assuring quality and control of costs.

The horizontal parts to an otherwise vertical set of parts enables the organization to be knit into a fabric instead of being loose strings. The horizontal integration is enabled by cross-functional committees which comprise of five or six members of top management, for each committee. The committee is responsible for establishing directions and policies for the function (eg., quality). It does not concern itself with the day to day management.

In the 1990s, Business Process Reengineering (BPR), a method evolved in the U.S. for overthrowing deficient processes an replacing them with efficient processes meets customer requirements more effectively has gained ground. BPR works closely with an advanced use of information technology. BPR has also emphasized the fact that processes are horizontal while organizations tend to be vertical. It has attempted to break the vertical structure down and replace it with horizontal designs. Such structures are called process based or team-based organizations. They enable the jobs of individuals to be enriched across departments, and clarify the flow of the processes towards the satisfaction of customers. This brings about job satisfaction. Cross functional management, Japanese style, will still need to be applied in order to integrate the functions of Q,C,D,S and M, but then the processes themselves manage Q,C, D,S,M in a more integrate the manner.

8.13 HYPOTHESIS

A hypothesis is a statement or proposition that is formulated to explain a certain phenomenon or set of observations. It is an educated guess or prediction about the relationship between variables that can be tested through empirical research.

In scientific research, hypotheses are used to guide the research process by providing a framework for collecting and analyzing data. Hypotheses can be either directional, which predict the direction of the relationship between variables (e.g., "increasing X will lead to an increase in Y"), or non-directional, which predict the existence of a relationship between variables (e.g., "there is a relationship between X and Y").

Once a hypothesis has been formulated, researchers can then design experiments or studies to test the hypothesis. The results of these experiments can either support or reject the hypothesis, which can then be used to refine the hypothesis or to develop new hypotheses for further research.

Hypothesis in TQM

In Total Quality Management (TQM), a hypothesis is a proposed explanation for a problem or challenge that a business is facing. The hypothesis is used to guide the development of solutions and to determine the effectiveness of those solutions through data analysis and testing.

TQM involves a continuous improvement process, where the organization identifies areas for improvement and uses data-driven analysis to develop solutions. The hypothesis helps to focus this process by identifying the potential causes of the problem and suggesting a solution that can be tested.

For example, if an organization is experiencing a high rate of defects in their products, a hypothesis might be that the defects are caused by a particular manufacturing process. The organization can then test this hypothesis by collecting data on the manufacturing process and analyzing it to determine whether there is a correlation between the process and the defects.

If the data supports the hypothesis, the organization can then develop and implement solutions to improve the manufacturing process and reduce the rate of defects. If the data does not support the hypothesis, the organization can refine the hypothesis or develop a new one to guide further investigation.

The use of hypotheses in TQM helps organizations to focus their efforts on identifying and solving the root causes of problems, rather than just treating the symptoms.

Some examples of hypotheses that could be used in Total Quality Management (TQM) are listed below:

Hypothesis : The high defect rate in a product is due to a problem with the manufacturing process. Solution: Conduct a detailed analysis of the manufacturing process and identify any areas where improvements could be made to reduce the defect rate.

Hypothesis : The low productivity of a team is due to poor communication and lack of collaboration. Solution: Implement team-building activities and communication training to improve collaboration and communication among team members.

Hypothesis : Customer complaints are due to a lack of quality control in the production process. Solution: Implement quality control measures to identify and correct any defects in the production process before products are shipped to customers.

Hypothesis : High employee turnover is due to a lack of training and development opportunities. Solution: Implement training and development programs to help employees develop new skills and advance their careers within the organization.

Hypothesis : Late delivery of products is due to poor supply chain management. Solution: Conduct an analysis of the supply chain and identify areas where improvements could be made to streamline the process and reduce delivery times.

These hypotheses are used to guide the development of solutions and to determine the effectiveness of those solutions through data analysis and testing in order to continuously improve the quality of products or services.

8.14 SUMMARY

Benchmarking, if carried out systematically, provides outside views and ideas and thus triggers off creativity of the group to strengthen its competitive capability. Benchmarking consists of analysis, comparison and synthesis. All these elements are present in all thetypes of benchmarking except performance benchmarking. Omission of any of these elements may cause sub-utilization of the potentiality of the concept. For example, use of database for the best practices does not generate enough commitments of the personnel concerned as a full-scale study of benchmarking does. As organizations strive to compete in the highly competitive global economy, it makes them focus their attention on the customer. QFD links customer requirements of 'whats' with the appropriate engineering design characteristics or 'hows' so the voice of the customer is translated into product designs and specifications. Building the House of Quality is an important step in moving from customer requirements to production requirements. Proponents of QFD argue that it has several benefits. The study of organizations implementing QFD demonstrates that it offers product development teams the opportunity to achieve significant improvements over traditional product development practices. QFD creates an information intensive atmosphere where communication increases and ideas are freely exchanged. This has a positive impact on developing product concepts and devising designs that meet customer quality and performance objectives.

Reliability is concerned with the failure in the field where the product is in use. It may be made a part of a requirement statement, which is later verified by measurement and tests. An effective reliability programme requires close monitoring of field failure. Like reliability, robustness of a design is also an important element in quality. A product needs to withstand a certain amount of vagaries when it is in use in the field. Taguchi one of the Japanese Gurus, has developed an off-line quality control methodology, to evolve optimum specifications through statistical experiments of designs. There should be a proper encouragement to employees to pay attention to these aspects. Another quality Guru, Crosby, considers, 'Zero Defects as the goal for any quality improvement exercise. He emphasizes management's consistent campaign and encouragement to motivate people to work towards zero defects. He has advocated 14 steps as a part of the implementation of Zero Defects Programme. Incremental improvements which are normally done on a continuous basis may be an inadequate response to a situation when an organization needs to improve substantially. An initiative, called 'Business Process Reengineering' is normally undertaken to analyse the business and develop radically new ways to carry out its certain aspects of it. Substantial improvement in quality in terms of various dimensions are achieved through such reengineering initiatives

8.15 KEY WORDS

House of Quality : A tabular presentation that facilitates planning of methods to meet specified customer requirements.

Performance Benchmarking: A process of comparison of business results of several competing organizations.

Problem Solving Process : A procedure commencing with selection of a problem till implementation that is followed by a quality circle in solving each problem.

Process Benchmarking : A comparison and action planning made for a specified functional process.

Product Benchmarking : A process that compares a product/service, offered by an organization with the same marketed by the competitors.

Strategic Benchmarking : A process of comparison and learning in respect of a corporate/business functional strategy.

Business Process : An ordered set of activities by which value is added to inputs.

Reliability Engineering : A discipline to deal with failures in order to analyse, predict and improve reliability.

8.16 SELF-ASSESSMENT QUESTIONS

1. What is benchmarking? How does it differ from an ad-hoc comparison?
2. What are the different types of benchmarking in relation to objects being benchmarked?
3. Classify benchmarking based on the nature of organizations against which benchmarking could be done?
4. What is a process? How does it differ from a function?
5. What are the different kinds of processes?
6. How does benchmarking help an organization?
7. What is QFD? What are the benefits of its application?
8. Explain briefly how you construct a house of quality?
9. What is reliability? Can you define it precisely?
10. Explain the concept of six sigma.
11. What is the role of hypothesis in TQM?

8.17 FURTHER READINGS

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